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The relationship between organizational focus on AI, financial growth and sustainable development: Evidence from Europe

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ABSTRACT

This study examines the link between organizations' focus on AI and their environmental, social, and governance (ESG) score. Furthermore, this study examines the relationship between organizations' AI focus and financial performance, measured by return on assets (ROA) and Tobin's Q. This manuscript relies on observations from a balanced panel of data comprising 432 publicly listed companies headquartered in Europe. The sample excludes banks and insurance companies, given their distinct accounting, governance, and capital structure standards. The sample consists of observations spanning from 2015 to 2023. Observations are gathered from LSEG Data & Analytics. We conduct baseline regression models. To ensure rigor, we also applied Hausman tests, variance inflation factors (VIF), and several robustness checks. The present manuscript is grounded in the economic theory framework. The empirical findings indicate: I) a positive and significant association between organizations' AI focus and their environmental (b = 0.127***; $p = 0.001$) and social pillar scores (b = 0.072**; $p = 0.023$); II) a positive and significant link with financial performance (ROA: b = 0.094**; $p = 0.012$; TobinQ: 0.103*; $p = 0.051$) and; III) a positive but statistically insignificant relationship with governance pillar scores (b = 0.030; $p = 0.166$). The obtained results yield significant contributions to both theory and practice. Specifically, the obtained results clarify and reconcile previously heterogeneous findings in the literature. Furthermore, it emphasizes that an organizational focus on AI may contribute to advancing the United Nations Sustainable Development Goals, while simultaneously enhancing financial performance.

1. Introduction

Artificial intelligence (AI) has recently gained momentum as a powerful tool that can deeply transform how organizations operate, pursue strategy and innovate. AI is expected to significantly impact organizations, markets and economies worldwide (Agrawal et al., 2019a, 2019b; Appio et al., 2024; Li et al., 2025).

Generally speaking, AI is detailed as an information technology (IT) that allows machines to perform generative and cognitive functions. These include tasks traditionally associated with understanding, learning, and interacting (Feuerriegel et al., 2023; Gupta et al., 2024). Although research on AI has been ongoing for decades, its popularity has surged only recently. The growing interest in AI is largely driven by the increasing computational power. This improvement results from advances in digitalization, computational tools, and data processing capabilities (Gill et al., 2022; Xiao et al., 2025). The integration of digital systems, connecting machines, communication devices, and sensors, combined with the exponential growth of digitized data, has created new opportunities for organizations. These opportunities allow them to leverage insights through innovative applications aimed at improving operational efficiency (Corazza et al., 2024; Frasca et al., 2024; Gupta et al., 2020; Lanzolla et al., 2020; Leone et al., 2021; La Torre et al., 2023). In fact, research estimates suggest that AI could generate an economic impact surpassing several trillion dollars by 2030. This potential stems from its ability to process vast amounts of unstructured data and generate predictive insights (Mishra et al., 2022). AI rapidly gained significance among businesses all over the globe due to its recognized ability to shape future entities' competitiveness (Enholm

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et al., 2022; Olan et al., 2022). Advancements in artificial neural networks and machine learning can be applied across a wide range of industries. These tools support data-driven models that optimize both financial and sustainability performance (Kulkov et al., 2024; Mishra et al., 2022; Monteiro et al., 2022; Peltier et al., 2024; Shiyyab et al., 2023). In fact, empirical research bodies started to conceptualize how the use of AI instruments would positively impact firms' financial performance (Monteiro et al., 2022; Shiyyab et al., 2023). Given the growing importance of corporate sustainability performance, scholars have directed their attention to AI. Recent studies explore its potential influence on organizations' environmental, social, and governance (ESG) performance (Gupta et al., 2021; Kulkov et al., 2024; Li et al., 2025; Wang et al., 2024). Examining the relationship between organizations' AI focus and their ESG performance is likely to yield valuable insights. Such analysis can underscore the capability of AI tools to contribute to the United Nations' Sustainable Development Goals (SDGs) (Damoah et al., 2021; Di Vaio et al., 2020; Singh et al., 2024). Numerous empirical studies have established a connection between firms' ESG performance and their ability to contribute to the SDGs (Delgado-Ceballos et al., 2023).

Some preliminary findings highlight the transformative potential of AI in enhancing organizations' financial, environmental, and social sustainability performance. However, part of the scholarly literature presents conflicting results (Kulkov et al., 2024). Scholars postulate that the impact of AI on the aforementioned performance metrics may be negative or statistically insignificant (Hang and Chen, 2022; Hudson and Morgan, 2024; Pan and Nishant, 2023). Specifically, scholars underscore the negative effect AI technologies may have on companies' financial performance. AI instruments often require businesses to adapt their existing IT systems, leading to significant capital absorption rather than the generation of immediate financial benefits (Bock et al., 2020; Makarius et al., 2020; Tingbani et al., 2024). Additionally, companies must invest heavily in human resources to ensure employees acquire the necessary skills and competencies to effectively leverage AI tools (Mishra et al., 2022; Tingbani et al., 2024). Finally, academic research points to the availability and quality of data as a major obstacle that can hinder improvements in firms' financial performance (Neiroukh et al., 2024; Tingbani et al., 2024).

From a sustainability perspective, scholars underscore the negative effect AI technologies might have on organizations' environmental and social sustainability endeavors. AI systems often require high degrees of computational power, which translates to substantial energy, water consumption, and increased carbon emissions (Akter et al., 2024; Haefner et al., 2023; Li et al., 2025; Pan and Nishant, 2023). Moreover, AI-driven models are likely to impact lower-skilled jobs, thereby widening the socioeconomic divide while perpetuating discriminatory behaviors caused by biased algorithms trained on historical data (Courtland, 2018; Kumar et al., 2023; Ledro et al., 2023; Peters et al., 2020; Singh et al., 2024).

AI integration produces varied and complex outcomes, and a significant research gap remains. This gap offers avenues for future studies to explore how AI adoption influences both financial and sustainability-related performance at the organizational level. The literature on AI's connection to firms' financial and sustainability performance remains heterogeneous. This underscores the need for scholars to further examine these relationships (Holmström and Carroll, 2024; Omrani et al., 2022; Peters et al., 2020). This line of inquiry is especially pertinent in assessing corporate contributions to the United Nations SDGs (Damoah et al., 2021; Gupta et al., 2024; Holmström and Carroll, 2024; Omrani et al., 2022; Peters et al., 2020). While AI's financial impacts are somewhat explored in today's academic discourse, its link to ESG dimensions remains underexplored. To advance this evolving research area, scholars are encouraged to adopt quantitative, firm-level methodologies. Existing research has largely relied on qualitative frameworks or macroeconomic perspectives (Gupta et al., 2024; Holmström and Carroll, 2024; Mishra et al., 2022; Rammer et al., 2022). This study aims to fill these gaps by analyzing the impact of AI adoption on companies' financial performance, measured through return on assets (ROA) and Tobin's Q. It also investigates sustainability performance, using ESG scores as a proxy. To enable a more detailed understanding, the ESG dimension is further disaggregated into its environmental, social, and governance components. This approach aligns with emerging calls for a more nuanced understanding of AI's role in advancing corporate sustainability (Starks, 2023; Di Vaio et al., 2020).

This manuscript seeks to contribute to the identified research gaps and reconcile conflicting results. It does so by analyzing the relationships between firms' AI focus and their financial and sustainability performance. The present manuscript adopts economic theory as its theoretical underpinning (Marwala and Hurwitz, 2017). Specifically, this manuscript focuses on companies whose headquarters are located in Europe due to the geographical focus on financial and sustainability rationales (Emma and Jennifer, 2021; Wang et al., 2025). Once all inclusion and exclusion criteria had been applied, the final sample consisted of 432 publicly listed companies with observations spanning from 2015 to 2023. This study adopts a methodological approach based on balanced panel data. Several robustness checks were conducted to ameliorate the reliability and validity of the baseline regression models and to test for potential endogeneity issues. The obtained results underscore the following relationships: I) firms' AI focus has a positive and significant effect on ROA and Tobin's Q, underscoring AI's beneficial impact on financial performance (Mishra et al., 2022; Monteiro et al., 2022; Peltier et al., 2024; Shiyyab et al., 2023); II) organizations' AI focus has a positive and significant effect on companies' environmental pillar score, highlighting the strength of this relationship (Akter et al., 2024; Kumar et al., 2023; Wang et al., 2024); III) organizations' AI focus has a positive and significant effect on firms' social pillar score, thereby corroborating part of the academic literature (Akter et al., 2024; Kumar et al., 2023; Rahman et al., 2023); IV) firms' focus on AI has a positive and paradoxical effect on their governance pillar score. This underscores the limited relationship between governance structures and the adoption of AI tools and systems. However, this manuscript acknowledges and underscores the complex nature associated with firms' AI focus and their governance pillar score (Schneider et al., 2023; Wirtz et al., 2020).

This manuscript adds to the growing literature on AI by providing firm-level, quantitative evidence from a large European sample. While prior studies have mainly relied on qualitative approaches, this study adopts a balanced panel data methodology to investigate the dual impact of organizations' AI focus on financial and sustainability (ESG) performance. Furthermore, it disaggregates the ESG dimensions, thereby offering a more nuanced understanding of AI interaction with different aspects of corporate sustainability. In doing so, this study provides insights into the role of AI as a potential enabler of the United Nations SDGs. This manuscript carries both managerial and theoretical contributions. Specifically, it adds to the discourse on the importance of AI for firms' financial and sustainability performance. First, this study underscores the importance of further exploring and clarifying the relationships between firms' AI focus and their financial and sustainability performance. In doing so, it contributes to the pursuit of the United Nations SDGs (Damoah et al., 2021; Di Vaio et al., 2020 Gupta et al., 2024; Holmström and Carroll, 2024; Leone et al., 2021; Mishra et al., 2022; Rammer et al., 2022). This is particularly relevant given the heterogeneous results currently published in academic literature. Second, the obtained results offer actionable insights to companies seeking to enhance their financial and sustainability performance. This manuscript supports practitioners' belief that a focus on AI can improve firms' financial and sustainability performance. In turn, this contributes to sustainable development while advancing the SDGs (Damoah et al., 2021; Di Vaio et al., 2020; Gupta et al., 2023; Mishra et al., 2022; Wamba-Taguimdje et al., 2020). Third, this study underscores the importance of disclosing companies' dynamism to stakeholders. Such disclosure is crucial, as dynamism contributes to operational efficiency, innovation, and value creation. Fourth, this study underscores the potential relationship between firms' AI focus and their corporate social responsibility frameworks. Such alignment can help organizations meet and surpass regulatory standards while promoting long-term sustainability (for instance, SDGs objectives). Finally, this manuscript contributes to the academic literature intersecting on the effect innovative digital solutions, such as AI, might have on companies' financial and sustainability performance.

The paper is structured as follows. Section 2 details the theoretical underpinnings and framework informing this body of research. Furthermore, Section 2 develops the research hypotheses of this manuscript. Thereafter, Section 3 discusses the sample composition and methodological approach employed. Section 4 details the obtained results and all robustness checks. Then, Section 5 details the discussion of the results. Conclusively, Section 6 unravels the concluding remarks, theoretical and managerial contributions, and limitations of this piece of academic research.

2. Theoretical background

2.1. Theoretical underpinning

AI is described as a technological tool capable of replicating human cognitive behaviors. It encompasses a wide range of technologies, including machine learning, neural networks, and deep learning (Agrawal et al., 2019a, 2019b). Recent advancements and the diffusion of AI tools have sparked considerable attention within the business realm since its application spans multiple industries (Gupta et al., 2020; Lanzolla et al., 2020). AI tools help organizations reimagine their business processes and strengthen their data mining capabilities. They can also support the adoption of dynamic pricing techniques and streamline activities such as reporting, data analysis, and process auditing (Dwivedi et al., 2023; Feuerriegel et al., 2023; Gupta et al., 2024).

AI's influence on businesses extends beyond the areas detailed above. It can also be employed to foster supply chain optimization, support research and development processes, and enhance customer and inventory management (Azadi et al., 2023; Giarmoleo et al., 2024; Kar et al., 2022; Toorajipour et al., 2021). In fact, empirical evidence underscores several reported benefits of AI. These include greater lead generation efficiency, enhanced organizational resilience, increased productivity, and improvements in both financial and sustainability performance (Far et al., 2023; Modgil et al., 2022; Parteka and Kordalska, 2023). Nonetheless, despite the growing emphasis on the potential benefits associated with firms' focus on AI tools, empirical quantitative studies remain limited. Few have assessed its relationship with firms' broader operational efficiency, financial, and sustainability performance (Gupta et al., 2024; Holmström and Carroll, 2024; Mishra et al., 2022; Rammer et al., 2022). While previous research has begun to conceptualize AI as an enabler for firms' success, important gaps remain. In particular, it remains unclear whether firms' AI focus directly impacts a company's accounting and market-based variables, as well as sustainability performance. Exploring the relationship between a firm's AI focus and its sustainability performance can provide valuable insights. It may help clarify whether AI instruments support the pursuit of the SDGs objectives set by national and supranational entities (Damoah et al., 2021; Di Vaio et al., 2020). This study aims to fill these gaps. In doing so, the authors consider it relevant to adopt economic theory, as suggested by Mishra et al. (2022), in an attempt to enhance the reliability and validity of the examined constructs and models.

2.2. Economic theory and hypotheses development: Financial and sustainability performance

2.2.1. The impact of firms' AI focus on their financial performance

This manuscript draws insights from economic theory, thereby focusing on the relationship between new technological adoption and organizations' ability to capitalize on them by ameliorating their financial performance. Specifically, Zeira (1998) argues that the adoption of new technological instruments (such as AI) is driven by the technological capabilities that organizations have already established. These capabilities shape how firms produce and deliver goods and services (La Torre et al., 2023; Jammeli et al., 2023). These newly integrated technological tools enable firms to generate and deliver their target output more efficiently and effectively. This leads to enhanced profitability, reduced waste, improved product quality and safety, while also nurturing better working conditions for employees (Bock et al., 2020; Makarius et al., 2020; Miozza et al., 2024; Triberti et al., 2022; Tingbani et al., 2024). Additionally, the betterment of an organization's financial performance through the introduction of new technological innovations can further fuel the company's reinvestment strategy. This creates a self-financing cycle that fosters the pursuit of projects related to sustainability and the SDGs (Damoah et al., 2021; Di Vaio et al., 2020). Therefore, building on the foregoing theoretical foundation, we now turn to the scholarly literature that examines the operational and productivity benefits associated with firms' AI focus.

La Torre et al. (2023) and Mishra et al. (2022) demonstrate how AI adoption can positively impact firms' productivity, enhance operational efficiency, and create new job opportunities. The resulting increase in productivity scales organizational output and is likely to positively influence firms' financial metrics (such as ROA) as these reflect the overall profitability (Parteka and Kordalska, 2023). Furthermore, the aforementioned considerations suggest a positive effect between firms' focus on AI tools and their Tobin's Q. This metric captures the market's perception of a company's growth potential, particularly when enhanced productivity is driven by the introduction of innovative technologies such as AI (Li et al., 2025; Mariani et al., 2023). The foregoing proposition is further corroborated by academic literature. Specifically, AI has been linked to improvements in firms' sales and marketing efficiency (Bock et al., 2020; Makarius et al., 2020; Mishra et al., 2022; Tingbani et al., 2024). Academic literature suggests that AI enhances firms' customer engagement strategies and transforms sales processes, thereby strengthening their financial position (Huang and Rust, 2021; Kaplan and Haenlein, 2019). From an operational perspective, AI possesses the ability to generate accurate forecasts, enabling companies to actively improve their inventory management costs. This supports a better understanding of market demand and more efficient resource allocation (Abrokwah-Larbi and Awuku-Larbi, 2024; Huang et al., 2023; Pham et al., 2024). Scholarly research presents empirical evidence showing that AI can support marketing and pricing strategies, thereby streamlining processes and reducing the costs traditionally associated with these activities. As a result, firms can improve their cost efficiency and achieve a lower expenditure per unit of assets (Davenport and Ronanki, 2018). Similarly, firms' adoption of AI instruments is likely to influence the market's assessment of their growth potential. An organization's focus on new technological tools signals its capacity to innovate and drive future growth (Ameye et al., 2023; Ahmad et al., 2021; Du and Xie, 2021; Roberts and Candi, 2024; Vlačić et al., 2021).

Nonetheless, scholarly discourse underscores several challenges associated with firms' AI focus and its effect on financial performance. First, empirical research suggests that firms' AI focus may have a negative effect on their financial and market performance. Specifically, researchers postulate that AI adoption requires substantial capital investments, particularly to hire specialized employees with the necessary skill sets. These requirements may lead to higher labor costs, which can offset the benefits associated with AI (Huang et al., 2023; Spring et al., 2022; Yu et al., 2023). Moreover, empirical evidence highlights that AI must be effectively integrated into a firm's existing digital infrastructure and processes. The extent to which organizations can realize the benefits commonly attributed to AI largely depends on their ability to seamlessly embed these tools within their operational systems. It is also contingent on the level of resources they are willing to invest to fully reap those benefits (Adobor et al., 2021; Bibri et al., 2024; Cao et al., 2021; Dwivedi et al., 2023). Furthermore, AI capabilities are highly dependent on the quality of the data on which algorithms and models are trained. Therefore, companies' focus on AI instruments could exacerbate existing organizational biases, thereby dampening AI's impact on the firms' financial performance and growth potential (Moradi and Dass, 2022; Rana et al., 2022; Schneider et al., 2023).

The reported observations from the previous paragraphs underscore the complex relationship between AI's potential cost savings and its associated expenses. However, despite the heterogeneous empirical evidence within academic literature, this manuscript aligns with the stream of research suggesting a positive association between firms' AI focus and their financial performance. This stance is informed by the theoretical framework adopted and by the contextual focus of this study, which examines organizations operating in developed economies where digitalization and innovation investments are prominent. This mitigates some of the challenges emphasized in the literature regarding the risks and costs associated with AI adoption. Following scholarly articles, firms' financial performance is measured using the accounting-based variable ROA and the market-based measure Tobin's Q (further detailed and justified in Section 3 of this manuscript). Considering the empirical findings outlined and the adopted theoretical framework, this manuscript proposes the following hypotheses:

H1a. Organizations' increased focus on AI instruments is positively associated with their ROA.

H1b. Organizations' increased focus on AI instruments is positively associated with their Tobin's Q.

2.2.2. The impact of firms' AI focus on their sustainability performance

Empirical evidence indicates that firms' integration of AI tools can enhance environmental sustainability performance. The discussion below synthesizes the scholarly literature on how an organizational focus on AI influences environmental outcomes and summarizes the key empirical findings. From an environmental perspective, scholarly articles indicate that a focus on AI innovation can help firms optimize resource usage, thereby reducing waste and limiting their environmental impact (Ahmad et al., 2021; Appio et al., 2024; Vaio et al., 2020; Nishant et al., 2020; Sipola et al., 2023). Empirical findings in the water management sector highlight the usefulness of AI instruments in supporting monitoring activities. These tools contribute to the efficient management of water resources and help maintain and improve water quality (Kar et al., 2022; Khakurel et al., 2018; Nishant et al., 2020; Omrani et al., 2022). However, the foregoing insights are not limited to the water management industry. Academic research underscores how the use of AI tools and innovations benefits the operational processes of agriculture, transportation, manufacturing, and many other sectors (Giordino et al., 2025a, 2025b; Giordano and Fantoni, 2025; Kar et al., 2022). Specifically, AI plays a significant role in advancing autonomous vehicles, which can lead to reduced emissions through models that optimize routes and driving conditions (Kar et al., 2022). Furthermore, researchers highlight how AI is being implemented to enhance the operational efficiency of manufacturing companies, thereby minimizing their ecological burden on natural resources (Ronaghi, 2023). AI can also be employed to enable or strengthen business models associated with recycling and circular practices (Ertz et al., 2022; Madanaguli et al., 2024; Ronaghi, 2023; Sjödin et al., 2023). Manufacturing firms' focus on AI is likely to improve their products and extend the manufacturing lifecycle, thereby enhancing environmental sustainability. When considering the broader context, AI-related technological innovations affect a wide range of stakeholders, including suppliers, distributors, and policymakers. These dynamics significantly reshape the logics and operations across the entire value chain. Investments in AI technologies are therefore likely to enhance firms' environmental sustainability, as a more efficient value chain reduces operational failures and disruptions while simultaneously improving output quality (Dwivedi et al., 2023; Mishra et al., 2022). Nonetheless, part of the scholarly literature challenges the findings outlined above, emphasizing numerous operational and organizational difficulties associated with AI adoption. First, the integration of AI models is linked to high levels of energy consumption, which is still predominantly derived from fossil fuels (Brevini, 2020; Kaplan and Haenlein, 2019; Kaplan and Haenlein, 2020). Second, the computational power needed to run AI algorithms and models relies on digital infrastructures built with rare metals and other materials, whose extraction depletes natural resources and harms the biosphere (Kaplan and Haenlein, 2020).

This paragraph examines the relationship between firms' AI focus and its impact on their social sustainability performance, as discussed in the scholarly literature. Academic writings explore and conceptualize the impact of AI on firms' social sustainability. The application of AI algorithms across various industry sectors can improve employees' working conditions, enhance product safety and security, and promote diversity and inclusion in the workplace (Chiarello et al., 2024; Giordano et al., 2024; Kar et al., 2022; John-Mathews, 2022; Nishant et al., 2020). Furthermore, some scholars suggest that AI models may positively influence workers' career development, as their adoption is often accompanied by additional training programs. However, part of the scholarly literature challenges the notion outlined above, as AI integration within organizations is often associated with job displacement (Du and Xie, 2021; Ertemel et al., 2021). This issue is particularly relevant for low-skilled jobs, where firms' AI focus may widen the gap between lower and higher-income populations. Additionally, since AI models and algorithms are trained on past observations, firms' reliance on them could reinforce and perpetuate discriminatory biases (Kar et al., 2022; Stahl et al., 2021).

This paragraph examines the relationship between firms' AI focus and its impact on their governance performance, as detailed in the scholarly literature and supported by empirical evidence. From a governance sustainability perspective, the academic discourse presents empirical evidence supporting several key propositions. Firms' focus on AI instruments is likely to strengthen corporate governance mechanisms, as these technologies can enhance transparency, accountability, and decision-making processes (Chhillar and Aguilera, 2022; Kar et al., 2022; Schneider et al., 2023; Tian et al., 2023). Specifically, firms can use AI to automate reporting procedures, improve fraud detection, and develop big data analytical capabilities that ground managerial decisions in factual observations. Empirical evidence has also begun to conceptualize how firms' adoption of AI technologies contributes to improved governance practices (Kar et al., 2022; Tian et al., 2023). However, some scholarly articles exploring firms' focus on AI and corporate governance suggest a more nuanced and complex relationship between the two. AI integration to support governance efforts also presents numerous challenges and barriers (Kar et al., 2022; Li et al., 2025). First, businesses must actively engage both management and employees with these new tools and systems. Otherwise, the organization's ability to benefit from enhancements, such as improved accountability and traceability within the governance structure, could be significantly hindered (Ahdadou et al., 2024). Second, organizations must carefully assess the technological literacy of their managers and directors. Insufficient expertise and understanding could significantly hinder the effective integration of AI technologies in support of organizational and governance mechanisms (Nishant et al., 2020). Third, an organization's governance and ownership structure, size, and culture strongly influence its ability to benefit from AI instruments in support of governance mechanisms (Heilinger et al., 2024; Schwaeke et al., 2025). Fourth, the external context of an organization (including market conditions, regulatory norms and frameworks, sociocultural contexts, and the influence of institutional investors) is highlighted in the scholarly discourse as another factor that can strongly impact the relationship between AI and the organization's governance pillar (Schwaeke et al., 2025). Fundamentally, part of the scholarly literature underscores the ambiguity of AI's benefits for organizational governance mechanisms and systems. Simultaneously, it highlights the complex and disruptive challenges associated with its integration (Ahdadou et al., 2024;

Schwaeke et al., 2025; Xiao et al., 2025).

Overall, based on the foregoing scholarly writings and the adopted theoretical underpinnings, the authors of this manuscript support the strand of literature arguing for AI's positive influence on companies' sustainability. This stance is informed by the theoretical framework adopted and by the contextual focus of this study. It examines organizations operating in developed economies, where sustainability principles are deeply embedded in the logics of both stakeholders and shareholders, as well as in national and supranational normative and strategic directions. Numerous scholarly articles underscore Europe's strong emphasis on sustainability issues and good governance practices within organizational environments (Giordino et al., 2024; Giordino et al., 2025a, 2025b; Truant et al., 2024; Velte, 2017; Wang et al., 2025). These logics are particularly prominent in publicly listed firms, given their obligation to comply with various regulations and social contracts. This context helps to mitigate some of the contradictory findings highlighted in the previous paragraphs. Following academic literature, we conceptualized sustainability through firms' ESG performance scores, which are likely to yield valuable insights into AI's capacity to foster the achievement of the United Nations SDGs (Damoah et al., 2021; Di Vaio et al., 2020; Vinuesa et al., 2020). Based on the above detailed notions, we conceptualize sustainability within the ESG performance dimension and propose the following hypotheses:

H2a. Organizations' increased focus on AI tools is positively associated with their environmental (E) pillar score.

H2b. Organizations' increased focus on AI tools is positively associated with their social (S) pillar score.

H2c. Organizations' increased focus on AI tools is positively associated with their governance (G) pillar score.

As outlined in the reported hypotheses, we have formulated each hypothesis for the individual dimensions of an organization's ESG score, adopting a more granular approach. Fig. 1 below provides a visual representation of our conceptual model.

3. Methodology

We employ fixed-effects panel regressions because they are particularly well-suited for quantitative studies estimating relationships among variables (Giordino et al., 2024; Giordino et al., 2025a, 2025b; Velte, 2017), such as firms' focus on AI and their financial and sustainability performance. This approach controls for unobservable, time-invariant firm characteristics that could otherwise confound the estimated relationships. Furthermore, by exploiting within-firm variation over time, fixed-effects regressions enhance the robustness and validity of the analysis, thereby providing more reliable insights into the impact of AI focus on firm performance.

3.1. Sample composition

The sample used in this study consists of observations gathered from LSEG Data and Analytics, a database widely employed in academic research.

When constructing the sample for this study, the authors applied specific inclusion and exclusion criteria. First, the sample focuses on firms headquartered in Europe. This decision is justified by Europe's strong emphasis on sustainability and reporting practices, which facilitates the observation of the effects between organizations' AI focus and its impact on their financial and sustainability performance (Emma and Jennifer, 2021). Second, all selected organizations have a market capitalization equal to or exceeding 250 million dollars. Consistent with previous academic research, this threshold was chosen to ensure the inclusion of organizations with publicly available, detailed, and reliable disclosures, which are essential for consistent data collection and analysis (Lins et al., 2019). While this inclusion criterion limits the scope of

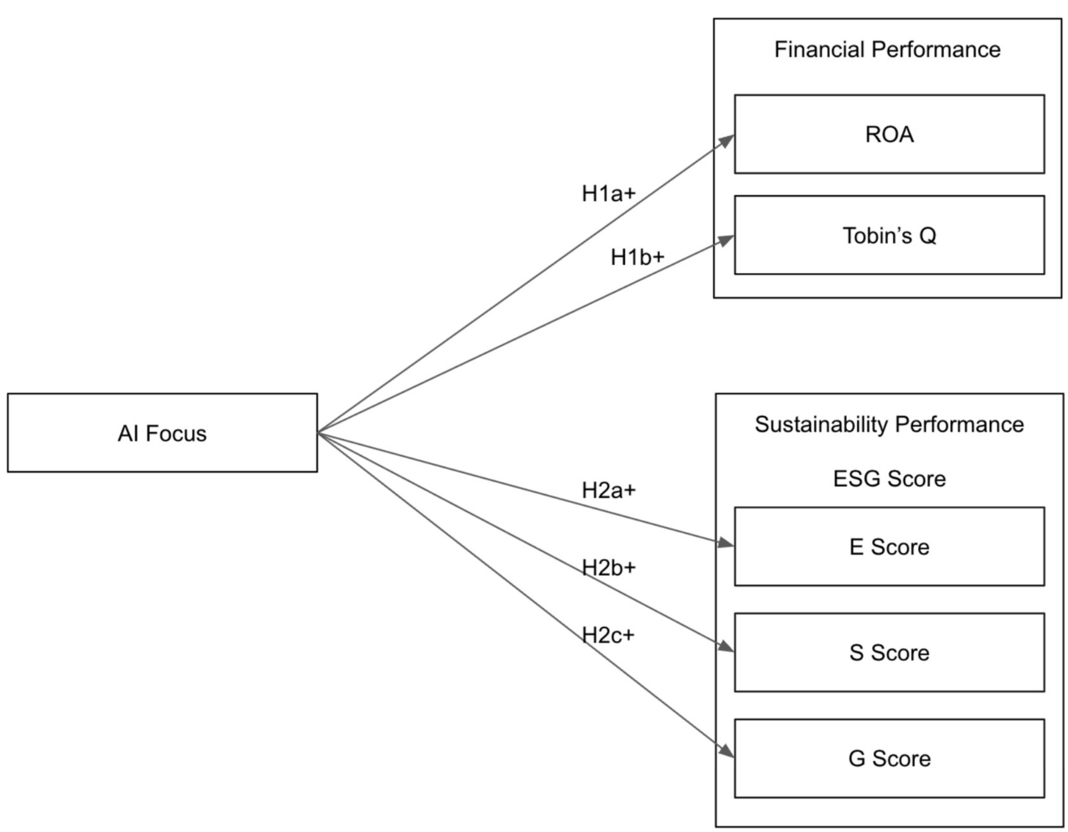

Fig. 1. Visual conceptual model (Authors' own elaboration).

the study to large-cap organizations, it enables a focus on companies with more standardized reporting practices. Furthermore, concentrating on large-cap organizations allows for insights from entities that are more likely to face stakeholder pressure to integrate AI into their operations and systems, thereby yielding more meaningful results (Kulkov et al., 2024; Li et al., 2025; Pan and Nishant, 2023). Third, the sample excludes banks and insurance companies, as they follow distinct accounting standards, governance structures, and capital requirements, which could otherwise distort the obtained results (Giordino et al., 2024). Fourth, all companies lacking data on their financial (ROA and Tobin's Q) and sustainability (ESG) performance for the selected time period were removed. Observations began in 2015 because the integration of AI models into corporate processes, strategies and reporting practices started gaining substantial traction that year. Finally, we excluded observations with incomplete or missing data; no imputation was performed. The shortlisted sample comprises 432 organizations and 3.456 observations spanning from 2015 to 2023. Table 1 presents the geographical and industrial distribution of the sample. Fig. 2 provides a graphical representation of the geographical distribution of the shortlisted companies.

3.2. Variables description

First, this article utilizes LSEG Data and Analytics' ESG as a proxy for companies' sustainability performance in relation to their ability to pursue SDG-related matters (Bekaert et al., 2023; Damoah et al., 2021). The authors chose LSEG Data and Analytics as their source of organizational ESG data since it is widely regarded as a clear, transparent, and objective indicator of a firm's ESG performance (Dobrick et al., 2023; Erhart, 2022; Velte, 2017). Specifically, in constructing its ESG assessment, LSEG Data and Analytics draws insights from publicly available data across a wide array of primary categories, as detailed in Table 2 of this manuscript (London Stock Exchange Group, 2024). LSEG Data and Analytics employs a weighting system to adjust the relative importance of each component of its ESG score, depending on the industry sector in which a company operates. Therefore, the authors of this study selected the LSEG Data and Analytics ESG score because it details and reports the weighting methodology employed, thereby enhancing the replicability and transparency of their observations (London Stock Exchange Group, 2024).

This manuscript employs each dimension of ESG to evaluate the impact of its components and to provide a more granular and comprehensive understanding of the relationship between firms' AI focus on their environmental (E), social (S), and governance (G) scores. In doing so, it seeks to uncover potential underlying drivers (Starks, 2023) while also offering more insightful perspectives on companies' performance and their contribution to achieving the United Nations' SDGs (Bekaert et al., 2023; Damoah et al., 2021).

Second, the manuscript employs ROA and Tobin's Q as proxies for firms' financial performance, thereby following the suggestions and guidelines of previous empirical studies (Choi and Wang, 2009; Velte, 2017). Specifically, ROA is considered one of the most widely used accounting-based variables to represent a company's financial performance, as it focuses on profitability relative to total assets. However, accounting-based variables are often influenced by earnings management practices (Choi and Wang, 2009). Therefore, to address the above-mentioned issue, the authors justify the inclusion of a market-based variable (Tobin's Q), thereby aligning their methodological approach with previous academic discourse (Choi and Wang, 2009; Velte, 2017). Tobin's Q is defined as the ratio between a firm's market value of physician assets and their replacement value. In this manuscript, we follow the approach of Choi and Wang (2009), comparing an organization's market value of equity and liabilities with its book value.

The following formulas have been adopted:

$$Return\ on\ Assets = \frac{Net\ Income\ before\ Preferred\ dividends + ((Interest\ expense\ on\ debt - interest\ capitalized) \times (1 - tax\ rate))}{Average\ of\ last\ year's\ and\ current\ year's\ total\ assets}$$

$$Tobin's\ Q = \frac{Market\ value\ of\ equity\ and\ liabilities}{Book\ value\ of\ equity\ and\ liabilities}$$

To evaluate the impact of firms' AI focus, the authors apply a one-year lag on the companies' ROA, Tobin's Q, and ESG score, since AI usage is unlikely to generate immediate improvements in financial returns (Velte, 2017).

Third, the authors developed a variable to measure organizations' focus on AI. The rationale is grounded in the notion that innovation activities are often reflected in corporate reporting. Companies seek to signal their technological advancements and strategic positioning to investors, the broader market, and other stakeholders (Ghosh et al.,

Table 1
Geographical and industry sector distribution of the shortlisted companies. Data from LSEG Data and Analytics. Authors' own elaboration.

Location			*GICS Industry*		
Headquarter location	*Frequency*	*Percentage*	*Industry Sector*	*Frequency*	*Percentage*
Austria	*1*	*0,23 %*	*Consumer Discretionary*	*106*	*24,54 %*
Belgium	*7*	*1,62 %*	*Communication Services*	*11*	*2,55 %*
Denmark	*3*	*0,69 %*	*Consumer Staples*	*52*	*12,04 %*
Finland	*8*	*1,85 %*	*Energy*	*32*	*7,41 %*
France	*68*	*15,74 %*	*Health Care*	*33*	*7,64 %*
Germany	*90*	*20,83 %*	*Industrials*	*138*	*31,94 %*
Ireland	*28*	*6,48 %*	*Information Technology*	*20*	*4,63 %*
Italy	*25*	*5,79 %*	*Materials*	*17*	*3,94 %*
Luxemburg	*9*	*2,08 %*	*Real Estate*	*2*	*0,46 %*
Netherlands	*23*	*5,32 %*	*Utilities*	*21*	*4,86 %*
Norway	*13*	*3,01 %*			
Spain	*7*	*1,62 %*			
Sweden	*18*	*4,17 %*			
United Kingdom	*132*	*30,56 %*			
Total	*432*	*100,00 %*	*Total*	*432*	*100,00 %*

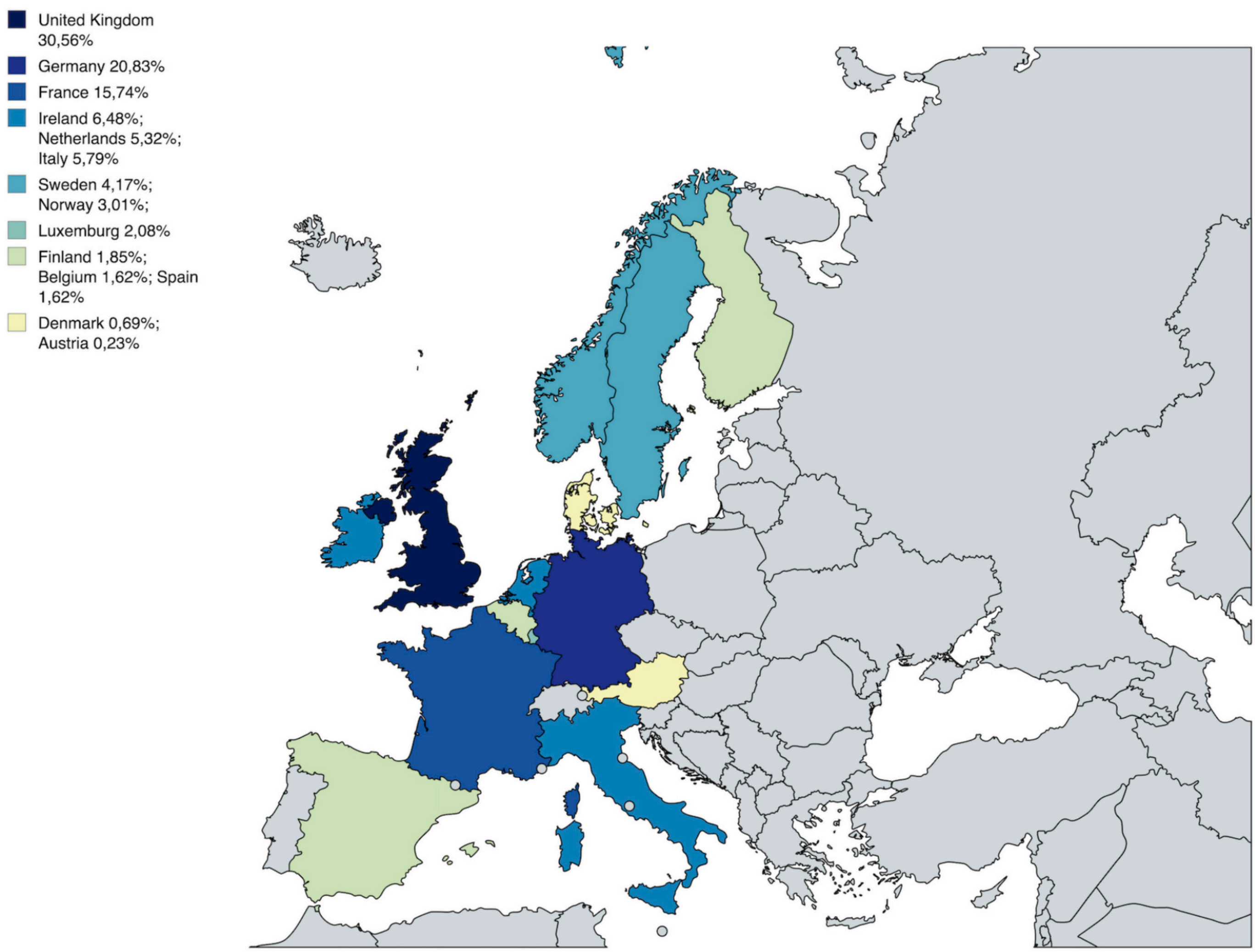

Fig. 2. A graphical representation of the geographical distribution of the shortlisted companies. Authors' own elaboration.

2025; Mishra et al., 2022). Accordingly, textual analysis of corporate disclosures can provide meaningful insights into a firm's operational focus and strategic orientation (Mishra et al., 2022; Rammer et al., 2022). Prior research supports the reliability and validity of this method, highlighting the informative value of firm-level reporting practices (Buehlmaier and Whited, 2018).

Following the methodology established by Mishra et al. (2022), the authors constructed the AI focus variable by analyzing firms' textual disclosures and identifying the occurrence of 122 terms associated with AI (Table 1 of the Appendix). The selection of these terms was informed by a qualitative review of corporate documents. The final variable was computed by counting the frequency of the identified terms within each firm's disclosure documents and normalizing this count by the total number of words in the analyzed text. The specific formula used is consistent with the approach proposed and detailed by Mishra et al. (2022):

$$AI\ Focus_{it} = \frac{Number\ of\ Words\ Related\ to\ AI_{it}}{Number\ of\ Words\ in\ their\ Reports\ and\ Statements} \times 100$$

In line with the approach adopted by Mishra et al. (2022), the authors validated the constructed measure of organizational focus on AI through a two-step procedure. First, we utilized two financial indicators. R&D expenditures reflect internal investments in AI-related innovation, while selling, general, and administrative (SGA) expenses may capture the outsourcing of AI capabilities via vendors. Second, we sorted the final sample into quintiles based on the level of AI focus. For each quintile, we calculated the average R&D and SGA expenditures, both normalized by total assets. As anticipated, firms in the lowest AI focus quintile reported significantly lower scaled R&D and SGA values compared to those in the highest quintile. The outcomes of this validation process are summarized in Table 2 of the Appendix. The resulting *p*-values are statistically significant at the 1 % level in most cases, reinforcing the notion that firms in the upper quintiles exhibit greater expenditures on R&D and SGA compared to those in the lower quintiles.

Given that our AI focus measure aggregates 122 AI-related terms, it is essential to identify which of these terms contribute most to the observed differences in R&D and SGA expenditures between firms with low and high AI focus. To this end, we rely on a bootstrap approach applied to the analysis detailed above, thereby assessing each AI term individually. We computed the mean differences in R&D and SGA spending between the first and fifth quintiles for 66 and 68 terms, respectively. For the remaining 56 (R&D) and 54 (SGA) terms, the limited variation across firms made comparison impossible. Among the analyzable terms, 8 (R&D) and 18 (SGA) showed no statistically significant differences in their means. This univariate examination indicates that approximately 52 % of AI terms are associated with significant differences in R&D spending, while 31 % are linked to differences in SGA.

Fourth, the authors employ various control variables, as well as industry, country, and year fixed effects to address potential endogeneity

Table 2
The following table details the categories and themes that LSEG Data and Analytics employs to calculate its ESG score (London Stock Exchange Group, 2024). Authors' own elaboration.

Variable	Pillars	Categories	Themes
ESG	Environmental	Emission	Biodiversity Emissions (GHG) Environmental management systems Total waste
		Innovation	Green revenues, R&D and capital expenditure Product innovation
		Resource use	Energy usage Environmental supply chain Sustainable Packaging Water usage
	Social	Community	
		Human rights	Human rights policies
		Product responsibility	Data privacy Product quality Responsible marketing
		Workforce	Career development and training hours Diversity and inclusion Health and safety Working conditions
	Governance	CSR strategy	CSR strategy ESG reporting and transparency
		Management	Compensation Structure (independence, diversity, committees)
		Shareholders	Shareholder rights Takeover defenses

concerns (Li et al., 2018). Specifically, the following control variables are included: I) company size, as academic literature highlights its influence on the pressure and expectations exerted by stakeholders, which can significantly affect both financial and sustainability performance (Giordino et al., 2024; Velte, 2017); II) firm resources, which are likely to shape both the ability and willingness to engage with AI instruments, as well as the capacity to translate innovative tools into drivers of financial and sustainability performance. Therefore, the following variables are included to control for the foregoing notion: property plant and equipment and debt ratio; III) the authors control for a company's liquidity using the ratio of cash dividends to total assets (Li et al., 2018); IV) research and development expenditure is employed, as prior scholars underscore its importance in innovation and sustainability endeavors (Kogut and Zander, 1992); V) the authors include a firm's beta as a control, since it measures the systemic risk of a company (Velte, 2017); VI) board gender diversity and board size are also employed, given that a company's governance structure strongly influences its ability to translate AI innovation into positive performance outcomes (Berger et al., 2014; Ghosh et al., 2025; Velte, 2017). Table 3 provides a detailed overview of all variables included in the models of this manuscript.

3.3. Baseline model design

The following section details the baseline models employed to test the hypotheses developed in Section 2 of this article. The authors used Lagrange Multiplier Tests and *F* tests to determine the appropriateness of panel data regressions. Moreover, the researchers applied Hausman Tests to determine whether fixed- or random-effects should be adopted. The results indicated that fixed effects were the appropriate choice for the models listed below. To address potential multicollinearity issues, the variance inflation factors (VIF) were also calculated. Since the obtained values did not exceed 4.75, it can be concluded that multicollinearity is unlikely to significantly affect the results (Hair, 2011).

Specifically, the authors employed the following equations:

Table 3
List of all the variables used in this manuscript, including a brief description, mean, and standard deviation. The sample covers 2015 to 2023. Authors' own elaboration.

Variable	Abbreviation	Description	Mean	Std. Dev.
E score	Escore	Environmental performance score generated by LSEG Data and Analytics	0.6669	0.3564
S score	Sscore	Social performance score generated by LSEG Data and Analytics	0.6927	0.3811
G score	Gscore	Governance performance score generated by LSEG Data and Analytics	0.4957	0.2331
ROA	ROA	Accounting-based variables focusing on a company's profitability in relation to its total assets	0.0893	0.0970
Tobin's Q	TobinQ	A ratio between an organization's market value of its equity and liabilities with its book value	2.8843	2.1124
AI Focus	AI	Firm's focus on AI technologies	0.0613	0.0198
Organization Size	Size	Natural logarithm of a firm's total assets	18.3431	3.9869
Property, plant and equipment	PPE	An organization's property, plant and equipment to total sales	0.4629	0.2947
Debt Ratio	Debt	A firm's observation detailing its debt to asset ratio	0.3851	0.3044
Liquidity	Liq	A firm's cash divided by its total assets	0.2008	0.0822
Research and Development	RD	An organization's R&D expenses divided by its revenues	0.2936	0.0735
Beta	Beta	Factor indicating the systemic risk associated to a specific entity	0.6825	0.1573
Board Gender Diversity	BoardGender	Variable indicating the ratio of a company's board gender diversity expressed as a percentage value	17.1211	14.7481
Board Size	BoardSize	Natural logarithm of a firm's size of its management board	11.5372	4.6295

$$1)\ ROA_{i,t} = \beta_0 + \beta_1 AI + \gamma Control_{i,t} + \Sigma Year_t + \Sigma Country_i + \Sigma Industry_i + \varepsilon_{i,t}$$

$$2)\ TobinQ_{i,t} = \beta_0 + \beta_1 AI + \gamma Control_{i,t} + \Sigma Year_t + \Sigma Country_i + \Sigma Industry_i + \varepsilon_{i,t}$$

$$3)\ Escore_{i,t} = \beta_0 + \beta_1 AI + \gamma Control_{i,t} + \Sigma Year_t + \Sigma Country_i + \Sigma Industry_i + \varepsilon_{i,t}$$

$$4)\ Sscore_{i,t} = \beta_0 + \beta_1 AI + \gamma Control_{i,t} + \Sigma Year_t + \Sigma Country_i + \Sigma Industry_i + \varepsilon_{i,t}$$

$$5)\ Gscore_{i,t} = \beta_0 + \beta_1 AI + \gamma Control_{i,t} + \Sigma Year_t + \Sigma Country_i + \Sigma Industry_i + \varepsilon_{i,t}$$

Fundamentally, $\gamma Control_{i,t}$ details the vector encompassing the control variables detailed in Table 3. Whereas, $\Sigma Year_t$, $\Sigma Country_i$ and $\Sigma Industry_i$ indicate the year-, country- and industry-fixed effects. Conclusively, $\varepsilon_{i,t}$ represents the error term.

Table 4
Presenting the results concerning Pearson's correlation matrix. Authors' own elaboration. The sample covers 2015 to 2023. Authors' own elaboration.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) AI	1													
(2) Escore	0.173*	1												
(3) Sscore	0.127	0.417*	1											
(4) Gscore	0.080	0.421*	0.501*	1										
(5) ROA	0.235*	0.373	0.097	0.266	1									
(6) TobinQ	0.181*	0.230	0.163	0.163	0.322*	1								
(7) Size	0.103	−0.234	0.183	0.198	0.205	0.404	1							
(8) PPE	0.090	0.072	0.080	0.207	0.269	0.333	0.222	1						
(9) Debt	−0.291	−0.099	−0.190	−0.137	−0.185	−0.229*	0.021	0.188	1					
(10) Liq	−0.401	−0.190	0.097	0.147	0.177	0.212	0.344	0.202	0.197	1				
(11) RD	0.059	0.084*	0.073	0.155	0.223*	0.194	0.298	0.212	0.224	0.251	1			
(12) Beta	0.204	−0.123*	−0.266	−0.122	−0.198*	−0.397*	0.220	0.195	0.033	0.212	0.339	1		
(13) BoardGender	0.055	0.093	0.117	0.204	0.092	0.117	0.187	0.152	−0.157	0.222	0.217	0.097	1	
(14) BoardSize	0.366	0.182	0.331	0.191	0.104	0.225	0.255	0.177	−0.189	0.131	0.228	0.174	0.221	1

* *indicates the level of significance of 5 % ($p < 0.05$).*

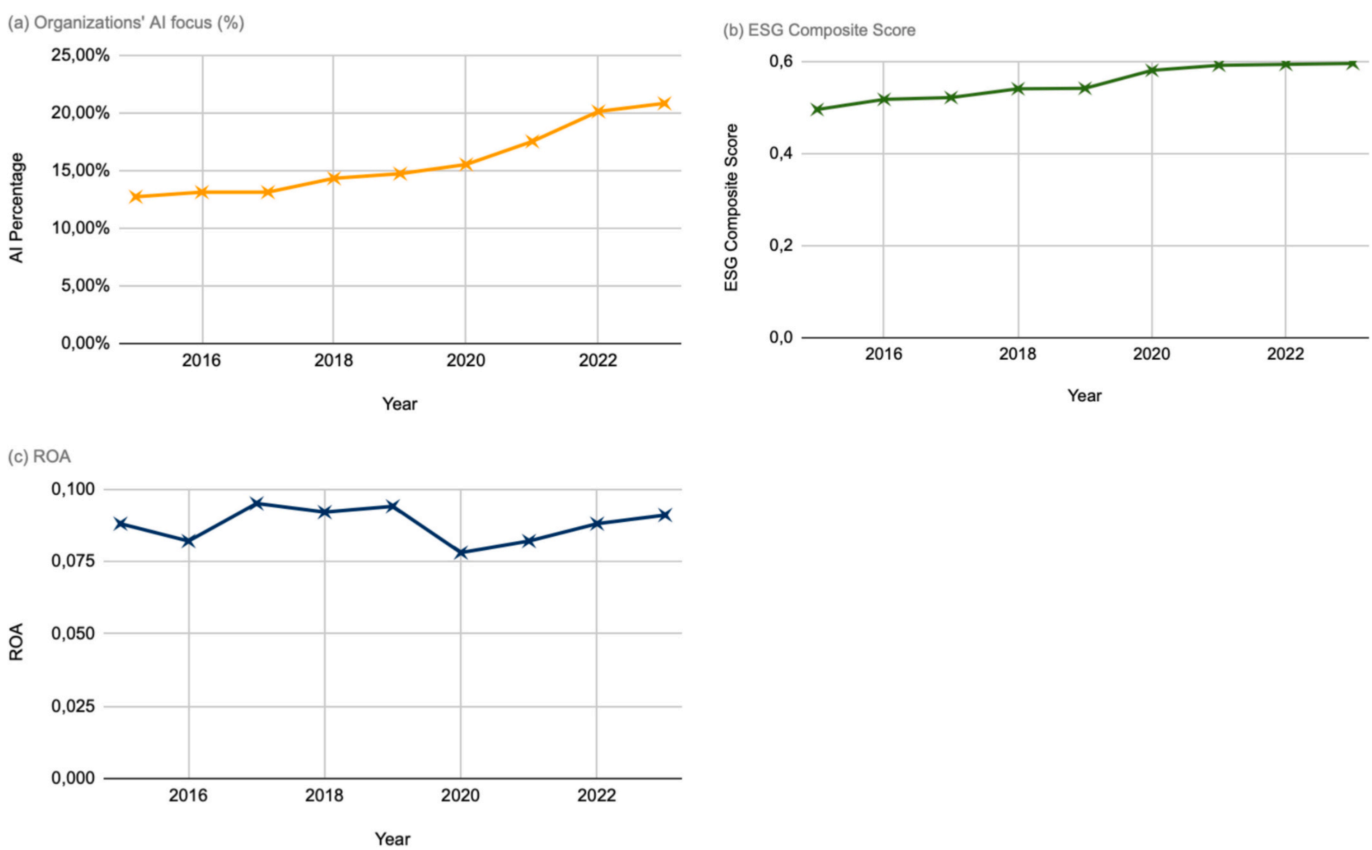

Fig. 3. Trends on average value of (a) Organizations' AI focus; (b) ESG composite score; and (c) ROA. Authors' own elaboration.

4. Results

4.1. Descriptive statistics

Table 3 presents the average values of the dependent, independent, and control variables, along with their standard deviations. On the other hand, Table 4 reports the results obtained from conducting a Pearson correlation matrix. The authors conducted the foregoing test to check for potential correlations between the variables employed in this manuscript (Velte, 2017). As depicted in Table 4, the obtained values do not raise specific concerns attributable to multicollinearity issues.

In an attempt to further detail the variable concerning organizations' AI focus, we have included in the Appendix (Table 3) the percentage distribution of the 122 selected AI-related terms, broken down by industry and year. Fig. 3 illustrates the trends concerning the following variables: organizations' AI focus, firms' ESG composite score, and their ROA. First, as depicted in Fig. 3, the organizations' AI focus, ESG composite score, and ROA show a steady rise reaching their peak during 2023. Second, the average value of ROA fluctuates, with its lowest value recorded in 2020.

4.2. Baseline regression

In the following section, the manuscript presents the results obtained from the regression models established in Section 3.3 of this article. Specifically, Table 5 presents the results obtained from the baseline regressions in this manuscript. As depicted in Table 5, the degree of R^2 and *Adjusted* R^2 is satisfactory.

Table 5
Results associated with the baseline regression analysis are detailed in Section 3.3 of this article. Authors' own elaboration.

	Model 1	*Model 2*	*Model 3*	*Model 4*	*Model 5*
Variables	*ROA*	*TobinQ*	*Escore*	*Sscore*	*Gscore*
AI	*0.094***	*0.103**	*0.127****	*0.072***	*0.030*
	(0.012)	*(0.051)*	*(0.001)*	*(0.023)*	*(0.166)*
Size	*0.013*	*0.007*	*0.013***	*0.017***	*0.009***
	(0.111)	*(0.121)*	*(0.041)*	*(0.041)*	*(0.038)*
PPE	*−0.007*	*0.006*	*−0.023**	*−0.022**	*−0.014**
	(0.108)	*(0.108)*	*(0.059)*	*(0.068)*	*(0.058)*
Debt	*−0.009*	*−0.020*	*−0.010****	*−0.009****	*−0.009****
	(0.113)	*(0.164)*	*(0.001)*	*(0.001)*	*(0.001)*
Liq	*0.065**	*0.070**	*−0.017*	*−0.014*	*−0.017*
	(0.054)	*(0.054)*	*(0.226)*	*(0.200)*	*(0.216)*
RD	*0.115**	*0.130**	*0.129*	*0.151*	*0.120*
	(0.072)	*(0.072)*	*(0.166)*	*(0.146)*	*(0.146)*
Beta	*−0.007**	*−0.009**	*−0.017***	*−0.009***	*−0.011***
	(0.063)	*(0.060)*	*(0.030)*	*(0.030)*	*(0.035)*
BoardGender	*0.020*	*0.077*	*0.151***	*0.163***	*0.122***
	(0.119)	*(0.119)*	*(0.41)*	*(0.041)*	*(0.040)*
BoardSize	*0.027*	*0.151**	*0.098**	*0.110***	*0.129***
	(0.112)	*(0.057)*	*(0.054)*	*(0.022)*	*(0.022)*
Observations	*3456*	*3456*	*3456*	*3456*	*3456*
R-squared	*0.228*	*0.233*	*0.230*	*0.205*	*0.238*
Adjusted R-square	*0.210*	*0.209*	*0.211*	*0.189*	*0.197*
Industry FE	*YES*	*YES*	*YES*	*YES*	*YES*
Country FE	*YES*	*YES*	*YES*	*YES*	*YES*
Year FE	*YES*	*YES*	*YES*	*YES*	*YES*

*This table presents the results from panel regression of organizations' AI focus (AI) on ROA (model 1), TobinQ (model 2), the environmental pillar score (Escore) (model 3), the social pillar score (Sscore) (model 4), the governance pillar score (Gscore) (model 5), and controls for the hole sample. This table details the results obtained through the baseline fixed effects regressions. Observations gathered from 2015 to 2023 were employed. The robust and standard errors are indicated in parentheses. Specifically, the reported p-values are two-tailed and the *, ** and *** symbols indicate significance at the 10 %, 5 % and 1 % levels, respectively.*

Model 1 indicates a positive and significant (b = 0.094; p = 0.012) relationship between a firm's AI focus and its ROA. This means that a 5 % increase in AI is associated with a 0.094 increase in ROA. Similarly, model 2 underscores a positive and significant relationship (b = 0.103; p = 0.051) between AI focus and TobinQ, suggesting that a 10 % increase in AI is associated with a 0.103 increase in TobinQ. Furthermore, model 3 results indicate the existence of a positive and significant (b = 0.127; p = 0.001) relationship between firms' AI focus and their environmental pillar score. This means that a 1 % increase in AI is associated with a 0.127 increase in EScore. Model 4 highlights the positive and significant relationship (b = 0.072; p = 0.023) between organizations' AI focus and their social pillar score, showing that a 5 % increase in AI is associated with a 0.072 increase in SScore. Finally, model 5 underscores the positive and non-significant relationship (b = 0.030; p = 0.166) between corporations' AI endeavors and their governance score. This means that while organizations with a stronger focus on AI may exhibit slightly higher governance pillar scores, the results obtained from our model cannot be considered statistically conclusive. This suggests that, within our sample, governance-related categories and themes are likely influenced by a broader set of institutional, regulatory, and cultural factors that extend beyond an organization's focus on technological investments, such as AI.

The reported results align with the hypotheses H1a, H1b, H2a, and H2b. However, when conducting the more granular examination of firms' AI focus on their ESG performance, the authors are able to confirm the proposed relationship of H2 with the exclusion of corporations' governance pillar score.

Table 6
Additional analysis includes only companies based in the UK and Ireland, conducted to explore potential differences between common and civil law countries. Authors' own elaboration.

	Model 6	*Model 7*	*Model 8*	*Model 9*	*Model 10*
Variables	*ROA*	*TobinQ*	*Escore*	*Sscore*	*Gscore*
AI	*0.088***	*0.093**	*0.111****	*0.068***	*0.027*
	(0.011)	*(0.050)*	*(0.001)*	*(0.021)*	*(0.146)*
Size	*0.010*	*0.011*	*0.015***	*0.020***	*0.006***
	(0.111)	*(0.125)*	*(0.040)*	*(0.040)*	*(0.037)*
PPE	*−0.004*	*−0.007*	*−0.019**	*−0.020**	*−0.014**
	(0.102)	*(0.102)*	*(0.053)*	*(0.061)*	*(0.052)*
Debt	*−0.009*	*−0.017*	*−0.012****	*−0.008****	*−0.012****
	(0.110)	*(0.166)*	*(0.001)*	*(0.001)*	*(0.001)*
Liq	*0.056***	*0.073**	*−0.017*	*−0.016*	*−0.016*
	(0.049)	*(0.057)*	*(0.221)*	*(0.208)*	*(0.211)*
RD	*0.109**	*0.122**	*0.131*	*0.146*	*0.118*
	(0.077)	*(0.077)*	*(0.149)*	*(0.148)*	*(0.148)*
Beta	*−0.006**	*−0.011**	*−0.014***	*−0.005***	*−0.012***
	(0.058)	*(0.055)*	*(0.033)*	*(0.033)*	*(0.038)*
BoardGender	*0.023*	*0.081*	*0.155***	*0.160***	*0.127***
	(0.121)	*(0.125)*	*(0.044)*	*(0.044)*	*(0.042)*
BoardSize	*0.023*	*0.149***	*0.090**	*0.118***	*0.121***
	(0.117)	*(0.061)*	*(0.057)*	*(0.030)*	*(0.031)*
Observations	*1280*	*1280*	*1280*	*1280*	*1280*
R-squared	*0.243*	*0.239*	*0.247*	*0.231*	*0.258*
Adjusted R-square	*0.237*	*0.217*	*0.225*	*0.219*	*0.231*
Industry FE	*YES*	*YES*	*YES*	*YES*	*YES*
Country FE	*YES*	*YES*	*YES*	*YES*	*YES*
Year FE	*YES*	*YES*	*YES*	*YES*	*YES*

*This table presents the results from panel regression of organizations' AI focus (AI) on ROA (model 6), TobinQ (model 7), the environmental pillar score (Escore) (model 8), the social pillar score (Sscore) (model 9), the governance pillar score (Gscore) (model 10), and controls for the hole sample. Observations gathered from 2015 to 2023 were employed. This analysis focuses on companies based in the UK and Ireland. The robust and standard errors are indicated in parentheses. Specifically, the reported p-values are two-tailed, and the *, ** and *** symbols indicate significance at the 10 %, 5 % and 1 % levels respectively.*

4.3. Robustness checks and additional analysis

The following section details the manuscript's additional analysis and robustness checks conducted to ensure that the baseline regression outcomes could be accepted.

First, the authors conducted a robustness check by replicating the baseline regressions using only companies that adhere to common laws. The foregoing decision is motivated by the need to test whether common and civil law systems yield differing results, thereby exploring how different geographical and cultural contexts can impact firms' AI focus effect on organizational financial and sustainability performance (de Jong and van der Meer, 2017).

As depicted in Table 6 of this article, the obtained results confirm the baseline regression outcomes. Specifically, models 6 and 7 confirm that firms' AI focus has a positive effect on companies' ROA (b = 0.088; p = 0.011) and TobinQ (b = 0.093; p = 0.050). This means that model 6 shows a 5 % increase in AI is associated with a 0.088 increase in ROA, while model 7 indicates a 10 % increase in AI is associated with a 0.093 increase in TobinQ. Additionally, models 8 and 9 corroborate the positive and significant relationship between organizations' AI focus and their environmental (b = 0.111; p = 0.001) and social (b = 0.068; p = 0.021) sustainability performance. Model 8 indicates that a 1 % increase in AI is associated with a 0.111 increase in EScore, whereas model 9 shows that a 5 % increase in AI is associated with a 0.068 increase in SScore. Finally, model 10 confirms the lack of a significant relationship between AI and organizations' governance pillar score.

Additionally, as suggested by Velte (2023), the authors employ the propensity score matching technique (PSM) to check for potential endogeneity concerns attributable to reverse causality and omitted

Table 7
Results obtained from the one-to-one PSM technique. Authors' own elaboration.

	Model 11	Model 12	Model 13	Model 14	Model 15
Variables	*ROA*	*TobinQ*	*Escore*	*Sscore*	*Gscore*
AI	0.073**	0.087**	0.103***	0.059**	0.029
	(0.011)	(0.041)	(0.001)	(0.019)	(0.136)
Size	0.009	0.005	0.009**	0.011**	0.007**
	(0.113)	(0.111)	(0.044)	(0.037)	(0.032)
PPE	−0.005	0.004*	−0.018*	−0.016*	−0.008*
	(0.102)	(0.103)	(0.052)	(0.062)	(0.051)
Debt	−0.007	−0.021	−0.008***	−0.007***	−0.007***
	(0.109)	(0.155)	(0.001)	(0.001)	(0.001)
Liq	0.058*	0.062*	−0.012	−0.011	−0.012
	(0.051)	(0.051)	(0.0217)	(0.191)	(0.206)
RD	0.108*	0.121*	0.122	0.144	0.114
	(0.067)	(0.063)	(0.154)	(0.139)	(0.140)
Beta	−0.005*	−0.007*	−0.012**	−0.007**	−0.009**
	(0.059)	(0.058)	(0.028)	(0.026)	(0.031)
BoardGender	0.017	0.074	0.147**	0.166**	0.116**
	(0.113)	(0.113)	(0.38)	(0.040)	(0.032)
BoardSize	0.022	0.147*	0.088*	0.103**	0.122**
	(0.105)	(0.057)	(0.051)	(0.027)	(0.019)
Observations	3456	3456	3456	3456	3456
R-squared	0.219	0.225	0.219	0.197	0.225
Adjusted R-square	0.209	0.217	0.208	0.191	0.217
Industry FE	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES

*This table details the results obtained through the PSM technique of organizations' AI focus (AI) on ROA (model 11), TobinQ (model 12), the environmental pillar score (Escore) (model 13), the social pillar score (Sscore) (model 14), the governance pillar score (Gscore) (model 15), and controls for the hole sample. Observations gathered from 2015 to 2023 were employed. The robust and standard errors are indicated in parentheses. Specifically, the reported p-values are two-tailed and the *, **, and *** symbols indicate significance at the 10 %, 5 % and 1 % levels respectively.*

variables (Rosenbaum and Rubin, 1983). Following the PSM approach, the authors divide the initial sample into two subsets composed of companies that belong to a geographical context characterized by I) low and; II) high organizations' AI focus. The resulting subsets are then denoted with a binary dummy variable. Thereafter, the researchers conduct a balanced test on the matched sample. The obtained results underscore no statistical disparities, thereby confirming the successful matching step previously carried out. Consequently, the authors re-estimate the baseline regressions using the matched sample developed through PSM, thus obtaining the results attributable to the technique. Specifically, Table 7 presents the obtained results from PSM.

As detailed in Table 7 of this manuscript, the PSM confirms the baseline regression results, thereby mitigating potential concerns attributable to endogeneity. Additionally, Table 4 of the Appendix reports the covariate balance test.

To further assess the validity and durability of the baseline regression models, we ran the models by including a two-year time lag between the independent, control and dependent variables. In doing so, we aimed to reduce the risks associated with endogeneity while adhering to business practices whereby the influence on organizations' sustainability (in this case, the ESG score) is lagged over time (Velte, 2023). As depicted in Table 8, the obtained results do further reinforce the baseline regression model. However, it is important to note that while the effects remain statistically significant, their magnitude declines slightly.

Furthermore, to test the robustness of our results, we re-estimated the initial models using MSCI ESG scores in place of LSEG Data & Analytics ESG scores. This substitution enabled us to further assess the validity of our proposed hypotheses and findings. While both organizations provide ESG assessments, their selected themes, key issues for each pillar, and overall methodologies differ slightly. Table 9 below details the obtained findings. The models detailed in Table 9 were run with a one-year time lag between the independent, control, and

Table 8
Results obtained from a two-year time lag to further test the obtained results validity and durability. The sample covers 2015 to 2023. Authors' own elaboration.

	Model 16	Model 17	Model 18	Model 19	Model 20
Variables	*ROA*	*TobinQ*	*Escore*	*Sscore*	*Gscore*
AI	0.064**	0.093*	0.099**	0.065**	0.033
	(0.021)	(0.062)	(0.017)	(0.033)	(0.144)
Size	0.005	0.006	0.008*	0.012*	0.006**
	(0.120)	(0.137)	(0.053)	(0.054)	(0.049)
PPE	−0.005	0.006	−0.019*	−0.015*	−0.009*
	(0.118)	(0.118)	(0.066)	(0.074)	(0.066)
Debt	−0.006	−0.018	−0.006***	−0.008***	−0.008***
	(0.130)	(0.173)	(0.001)	(0.001)	(0.001)
Liq	0.057*	0.066*	−0.013	−0.012	−0.014
	(0.060)	(0.060)	(0.244)	(0.210)	(0.225)
RD	0.101*	0.124*	0.117	0.140	0.113
	(0.088)	(0.085)	(0.187)	(0.155)	(0.160)
Beta	−0.005*	−0.007*	−0.011**	−0.009**	−0.008*
	(0.080)	(0.080)	(0.048)	(0.046)	(0.051)
BoardGender	0.012	0.063	0.138*	0.147*	0.109*
	(0.139)	(0.139)	(0.57)	(0.057)	(0.054)
BoardSize	0.019	0.138*	0.088*	0.095**	0.113**
	(0.122)	(0.066)	(0.054)	(0.038)	(0.040)
Observations	3456	3456	3456	3456	3456
R-squared	0.226	0.228	0.227	0.207	0.235
Adjusted R-square	0.213	0.211	0.211	0.194	0.203
Industry FE	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES

*This table presents the results obtained from a 2 year time lag of organizations' AI focus (AI) on ROA (model 16), TobinQ (model 17), the environmental pillar score (Escore) (model 18), the social pillar score (Sscore) (model 19), the governance pillar score (Gscore) (model 20), and controls for the hole sample. Observations gathered from 2015 to 2023 were employed. The robust and standard errors are indicated in parentheses. Specifically, the reported p-values are two-tailed, and the *, **, and *** symbols indicate significance at the 10 %, 5 % and 1 % levels respectively.*

Table 9
Results from the regression analysis using ESG pillar scores sourced from MSCI. The sample covers 2015 to 2023. Authors' own elaboration.

	Model 21	Model 22	Model 23
Variables	*Escore_MSCI*	*Sscore_MSCI*	*Gscore_MSCI*
AI	0.111***(0.001)	0.090**(0.031)	0.041 (0.133)
Size	0.022**(0.033)	0.021**(0.032)	0.015**(0.047)
PPE	−0.037**(0.048)	−0.011*(0.054)	−0.011*(0.062)
Debt	−0.021***(0.001)	−0.017***(0.001)	−0.015***(0.001)
Liq	−0.022 (0.264)	−0.011 (0.264)	−0.024 (0.258)
RD	0.133 (0.174)	0.140 (0.167)	0.137 (0.152)
Beta	−0.018**(0.034)	−0.014**(0.037)	−0.010**(0.037)
BoardGender	0.147**(0.38)	0.170**(0.036)	0.133**(0.031)
BoardSize	0.072*(0.061)	0.106**(0.028)	0.150**(0.019)
Observations	3456	3456	3456
R-squared	0.238	0.211	0.230
Adjusted R-square	0.220	0.203	0.207
Industry FE	YES	YES	YES
Country FE	YES	YES	YES
Year FE	YES	YES	YES

*This table presents the results from panel regression of organizations' AI focus (AI) on the environmental pillar score (Escore_MSCI) (model 21), the social pillar score (Sscore_MSCI) (model 22), the governance pillar score (Gscore_MSCI) (model 23), and controls for the whole sample. Observations gathered from 2015 to 2023 were employed. The robust and standard errors are indicated in parentheses. Specifically, the reported p-values are two-tailed, and the *, **, and *** symbols indicate significance at the 10 %, 5 %, and 1 % levels, respectively.*

dependent variables.

Finally, as an additional analysis, we explored the potential non-linear relationships between organizations' AI focus and their financial and sustainability performance. Specifically, ROA and TobinQ for

Table 10
Panel data regression results investigating the non-linear relationship. Authors' own elaboration.

	Model 24	*Model 25*	*Model 26*	*Model 27*	*Model 28*
Variables	*ROA*	*TobinQ*	*Escore*	*Sscore*	*Gscore*
AI	*0.075**(0.034)*	*0.063*(0.088)*	*0.127**(0.018)*	*0.046**(0.047)*	*0.220 (0.341)*
AI²	*0.002**(0.048)*	*0.004*(0.097)*	*0.005**(0.044)*	*0.003*(0.074)*	*0.002 (0.577)*
Size	*0.211 (0.335)*	*0.024 (0.327)*	*0.022**(0.022)*	*0.054*(0.077)*	*0.010*(0.068)*
PPE	*0.022 (0.142)*	*0.079 (0.417)*	*−0.003*(0.053)*	*−0.023*(0.059)*	*−0.013*(0.067)*
Debt	*−0.019 (0.204)*	*−0.014 (0.135)*	*−0.009***(0.001)*	*−0.012***(0.001)*	*−0.008***(0.001)*
Liq	*0.053*(0.068)*	*0.049*(0.068)*	*−0.118 (0.242)*	*−0.196 (0.290)*	*−0.154 (0.292)*
RD	*0.073*(0.060)*	*0.079*(0.080)*	*0.269 (0.221)*	*0.231 (0.242)*	*0.183 (0.234)*
Beta	*−0.008*(0.072)*	*−0.009*(0.069)*	*−0.015**(0.034)*	*−0.004**(0.035)*	*−0.013**(0.049)*
BoardGender	*0.053 (0.250)*	*0.088 (0.290)*	*0.150**(0.37)*	*0.143**(0.034)*	*0.121**(0.049)*
BoardSize	*0.059 (0.119)*	*0.127*(0.077)*	*0.099*(0.067)*	*0.108**(0.030)*	*0.131**(0.029)*

*This table presents the results from panel regression investigating the non-linear relationship of organizations' AI focus on ROA (model 24), TobinQ (model 25), the environmental pillar score (Escore) (model 26), the social pillar score (Sscore) (model 27), the governance pillar score (Gscore) (model 28), and controls for the hole sample. Observations gathered from 2015 to 2023 were employed. The robust and standard errors are indicated in parentheses. Specifically, the reported p-values are two-tailed and the *, **, and *** symbols indicate significance at the 10 %, 5 % and 1 % levels, respectively.*

financial performance, and Escore, Sscore, and Gscore for sustainability performance. Table 10 presents the results of the non-linear effects. It is important to note that, in order to establish a non-linear relationship, y must be regressed on x and x^2. Within this context, it is not appropriate to assess the impact of a change in x while x^2 is kept constant. Consequently, to account for it, $\frac{dy}{dx}$ must be computed accordingly (Wooldridge, 2009). Henceforth, $\frac{dx}{dy} = b_1 + 2b_2x$.

The non-linear specification detailed in Table 10 underscores the following. First, as the coefficients of both AI and AI^2 are positive, we conclude that AI and ROA, as well as TobinQ, exhibit a linear relationship. Second, as the coefficients of both AI and AI^2 are positive, we conclude that AI, Escore and Sscore exhibit a linear relationship. Finally, as depicted in Table 10, it becomes evident that AI does not have a significant association with Gscore.

5. Discussion

The manuscript results reveal that companies' AI focus has a positive and significant effect on their financial performance, thus allowing the authors to confirm hypotheses H1a and H1b. Therefore, the present academic article corroborates previous scholarly literature demonstrating a positive relationship between firms' AI focus and its effect on financial performance (Bock et al., 2020; Makarius et al., 2020; Mishra et al., 2022; Tingbani et al., 2024).

Specifically, the use of AI can positively impact organizations' productivity, operational efficiency and output scalability, thereby positively influencing their financial metrics such as ROA (Parteka and Kordalska, 2023). Moreover, firms' AI focus has a positive impact from a market perception perspective, as external stakeholders are likely to direct attention to the enhanced growth potential and capabilities of organizations employing AI instruments (Mariani et al., 2023). The obtained results further support the proposition made by the above scholarly writings, as they demonstrate a positive association between corporations' AI focus and their financial performance through a market metric such as Tobin's Q.

Additionally, the regression analysis employed aligns with the literature stream detailing the sales and marketing efficiencies associated with companies' adoption of AI technologies (Bock et al., 2020; Makarius et al., 2020; Mishra et al., 2022; Tingbani et al., 2024). In fact, AI can help companies improve their customer engagement approaches and revolutionize their sales processes, thus strengthening organizations' ability to generate higher financial returns (Huang and Rust, 2021; Kaplan and Haenlein, 2019).

The foregoing benefits are often attributable to AI's capability to generate accurate forecasts, thereby allowing firms to manage inventory more accurately and efficiently, while reducing the costs traditionally associated with inventory management (Pham et al., 2024). Furthermore, this empirical investigation supports and fuels the academic discourse that identifies AI predictive capabilities as beneficial to corporations' financial returns, owing to its ability to better predict market demands, thus allowing a more efficient resource allocation (Abrokwah-Larbi and Awuku-Larbi, 2024; Huang et al., 2023; Pham et al., 2024). Additionally, firms' AI focus has a positive impact on their ROA and Tobin's Q, as AI tools are likely to influence market perceptions and growth potential. An organization's focus on technological innovation signals its capacity to adapt and respond to change, which can enhance investor confidence and valuation (Ameye et al., 2023; Ahmad et al., 2021; Du and Xie, 2021; Ghosh et al., 2025; Roberts and Candi, 2024; Vlačić et al., 2021).

On the other hand, the obtained results contradict previous empirical studies that report the negative impact of AI instruments on companies' financial performance. Despite the need for a specialized workforce and additional costs associated with training and reskilling, organizations that focus on integrating AI tools are likely to experience financial growth (Huang et al., 2023; Spring et al., 2022; Yu et al., 2023). Moreover, the obtained results further contradict the literature which details the costs associated with implementing AI within already existing digital systems as prohibitively high and damaging to companies' financial performance (Adobor et al., 2021; Bibri et al., 2024; Cao et al., 2021; Xiao et al., 2025).

On the other hand, the obtained empirical findings support the academic literature detailing the positive influence of AI focus on companies' sustainability endeavors. In fact, hypotheses H2a and H2b can be confirmed, with the exception of hypothesis H2c concerning firms' governance pillar score. The more granular analysis further supports the academic writings that posit the positive effect of firms' AI focus on their environmental and social sustainability performance (Ahmad et al., 2021; Di Vaio et al., 2020; Nishant et al., 2020; Sipola et al., 2023). The integration of AI logics within organizations can help support and achieve the United Nations' SDGs, as nurturing environmental and social performance actively contributes to corporations' alignment with the SDGs (Bekaert et al., 2023; Damoah et al., 2021; Di Vaio et al., 2020).

Specifically, firms' AI focus impacts their operational performance, resulting in reduced waste and more efficient resource utilization, thereby limiting their environmental impact and footprint (SDG9, SDG12, SDG13, SDG14, SDG15) (Ahmad et al., 2021; Di Vaio et al., 2020; Nishant et al., 2020; Sipola et al., 2023). The reported benefits have been observed across various industry sectors, underscoring how AI applications are likely to benefit entities operating in diverse segments (as further corroborated by this manuscript's results) (Kar et al., 2022; Nishant et al., 2020). By minimizing resource waste, entities focusing their efforts on AI are likely to reduce their ecological burden on natural resources and the biosphere, thus positively contributing to the progress needed to achieve the SDGs, specifically, SDG14 and SDG15 (Bekaert et al., 2023; Ronaghi, 2023; Pan and Nishant, 2023). In fact, part of the academic literature identifies AI as a potential enabler for business models focusing on circular economy and recycling logics, thereby underscoring its importance for pursuing the SDGs (such as SDG9), as well as AI's capability to enhance companies' environmental and social performance (such as SDG8, SDG9, SDG14, and SDG15) (Ertz et al., 2022; Li et al., 2025; Madanaguli et al., 2024; Ronaghi, 2023; Sjödin et al., 2023).

The present manuscript does not support the academic discourse proposing a negative relationship between organizations' AI focus and their environmental performance. The negative environmental effects attributable to AI's consumption of energy and use of strategic and rare minerals do not negatively impact firms' environmental performance (Kaplan and Haenlein, 2019; John-Mathews, 2022; Kaplan and Haenlein, 2020; Omrani et al., 2022; Pan and Nishant, 2023).

From an environmental sustainability perspective, policymakers can draw on these insights to encourage companies to incorporate AI not only as a technological tool but also as a mechanism to strengthen ESG data quality, reporting consistency, and compliance with regulatory requirements. This aligns with the objectives of the European Corporate Sustainability Reporting Directive. Moreover, the evidence that AI can enhance environmental and social outcomes resonates with the European Green Deal and broader United Nations SDGs. This underscores the importance of integrating AI into public policy agendas to accelerate firms' contributions to climate neutrality, resource efficiency, and social well-being among many others. Sustainability disclosure under the European Corporate Sustainability Reporting Directive reduces information asymmetries and measurement costs, thereby improving price discovery and capital allocation, in line with economic theory and with the obtained evidence concerning organizations' AI focus and their financial performance, as well as environmental and social outcomes.

From a social sustainability perspective, the obtained results align with the academic discourse proposing a positive association between organizations' AI focus and their social sustainability performance (Kar et al., 2022; Nishant et al., 2020). Thereby, the integration of AI can improve employees' working conditions, enhance product safety and security while promoting diversity and inclusion (SDG5, SDG8, SDG10). Additionally, the integration of AI models is likely to benefit employees, as they may receive additional training (SDG4). Organizations' AI focus positively impacts their social sustainability performance, thus contributing to their pursuit of the United Nations' SDGs (Bekaert et al., 2023; Damoah et al., 2021; John-Mathews, 2022; Omrani et al., 2022; Pan and Nishant, 2023). By reinforcing the academic literature that presents empirical evidence corroborating the positive correlation between the two foregoing variables, this manuscript further strengthens the connection between AI usage and companies' pursuit of SDGs.

On the other hand, this manuscript does not support the academic literature postulating a negative association between AI and companies' social sustainability performance (Du and Xie, 2021; Ertemel et al., 2021; Li et al., 2025). Therefore, the proposed issues associated with job displacements, algorithmic biases (Courtland, 2018), and a widened gap between lower and higher income groups do not seem to negatively impact firms' sustainability pillar score (Kar et al., 2022; Stahl et al., 2021).

Finally, due to the positive and non-significant result for companies' governance pillar score, the present body of literature neither supports nor disputes previously published scholarly works. Nonetheless, it underscores the nuanced and complex relationship between organizations' AI focus and their governance pillar score (Ahdadou et al., 2024; Schwaeke et al., 2025), thereby prompting the need for further scholarly research to clarify the relationship between organizations' AI focus and their governance structures and pillars.

The forthcoming European AI Act provides a regulatory framework for governing the responsible adoption of AI, and our findings suggest that its implementation will be critical to ensuring that AI use strengthens rather than undermines sustainability goals. This helps internalize potential externalities and lower adoption uncertainty, increasing the expected net benefits of AI investment and reinforcing the empirical links in the literature between AI capability, firm productivity and valuation, and improved sustainability performance.

6. Conclusion

This empirical study explores the relationship between firms' AI focus and their financial and sustainability performance in an attempt to better comprehend AI's role in pursuing the United Nations' SDGs. Based on the previously discussed details and theoretical underpinnings, the authors adopt a balanced panel data fixed effects regression model. In doing so, they are able to respond to the research gap that calls for additional quantitative studies focusing on organizations' AI focus (Gupta et al., 2024; Holmström and Carroll, 2024; Mishra et al., 2022; Rammer et al., 2022). The sample used for this research comprises companies headquartered in Europe with a capitalization of more than 250 million dollars. Moreover, the sample does not include observations from banks and insurance companies due to their different accounting, governance, and capital structure standards (Giordino et al., 2024; Giordino et al., 2025a, 2025b). The gathered observations span from 2015 to 2023. Furthermore, the manuscript combines both baseline regression models and additional analyses to test the robustness and reliability of the obtained findings.

The obtained findings help to bridge the literature gap by providing additional empirical evidence on the impact of AI on firms' financial and sustainability performance. In doing so, the study connects the integration of AI with corporations' contribution to the SDGs (Damoah et al., 2021; Gupta et al., 2024; Holmström and Carroll, 2024). Specifically, the obtained results indicate a positive and significant relationship between firms' AI focus and their financial performance (expressed as ROA and Tobin's Q) and their environmental and social sustainability performance (detailed as E and S pillar scores of ESG performance). Finally, the authors identify a positive but non-significant relationship between firms' AI and their governance pillar score.

Below, the authors have outlined both theoretical and managerial implications, as well as limitations and avenues for future research.

6.1. Theoretical implications

First, this manuscript contributes to the academic literature focusing on the impact and evolution of AI instruments within the business and management field. Specifically, it explores the link between organizations' AI focus and their financial and sustainability performance (Damoah et al., 2021; Di Vaio et al., 2020; Gupta et al., 2024; Holmström and Carroll, 2024; Mishra et al., 2022; Rammer et al., 2022). In doing so, corporations are able to improve their environmental and social contributions, thereby fostering their sustainability positioning. Organizations can actively contribute to the pursuit of the United Nations' SDGs (Bekaert et al., 2023; Damoah et al., 2021). The foregoing contribution is particularly relevant for the academic discourse due to the heterogeneous findings reported within AI scholarly writings (Gupta et al., 2024; Mishra et al., 2022).

Second, this manuscript contributes to theory by conceptualizing and employing economic theory as its theoretical framework (Marwala and Hurwitz, 2017). In doing so, this study opens new avenues for future researchers to employ varying theoretical underpinnings to inform their research endeavors.

Finally, this paper contributes to the literature by further developing the academic discourse that connects firms' disclosure practices and their performance (Gupta et al., 2023; Mishra et al., 2022; Wamba-Taguimdje et al., 2020). The foregoing contribution is particularly relevant in the conceptualization of firms' ESG performance as an indicator of their ability to achieve SDGs. Moreover, the above-detailed notion is also evident in the conceptualization of firms' AI focus discourse and their corporate reports. The variable employed to measure AI within organizations heavily relies on these disclosures (Mishra et al., 2022).

6.2. Practical implications

First, the present study underscores the market interest in firms' AI focus through its analysis of AI effects on organizations' Tobin's Q (Mishra et al., 2022; Monteiro et al., 2022; Peltier et al., 2024; Shiyyab et al., 2023). It clearly communicates the importance that various external stakeholders place on companies' ability to innovate and successfully integrate new technologies (such as AI) into their business models and processes. This manuscript offers several actionable recommendations for managers and organizations. For example, managers aiming to minimize the negative market perception associated with the increased costs and expenses of AI adoption may need to consider the effect that an organization's AI focus has on its Tobin's Q. In doing so, they can build a more compelling case for continued investment in emerging technologies and tools, thereby reinforcing the strategic importance of technological advancements.

Second, this study underscores the importance of firms' AI focus in enhancing their financial performance due to AI's ability to improve their operational, marketing, forecasting, and inventory management efficiency (Peltier et al., 2024; Shiyyab et al., 2023). It emphasizes the importance for managers and board members to carefully assess the potential contributions of innovative technologies such as AI to the long-term success and development of their organization. The obtained results underscore the need for managers to assess the trickle-down effect of AI, thus highlighting the link that might exist between organizations' AI and ROA.

Third, the observations and variables used in this study underscore the need for organizations and managers to actively engage in innovation and sustainability narratives when constructing their reports. In doing so, entities can effectively communicate their strategic orientation while also underscoring their practices and instruments. However, it is important to note that innovations, such as AI, should be linked with organizations' sustainability performance to further emphasize their multifaceted approach and compliance with social contracts and institutional goals.

Finally, it highlights the positive impact that a firm's focus on AI can have on its environmental and social performance, thereby strengthening its ability to comply with increasingly stringent regulatory frameworks developed by national and supranational institutions worldwide (Akter et al., 2024; Kumar et al., 2023; Wang et al., 2024). Specifically, this article details AI's positive impact on firms' environmental and social pillar scores, thereby linking it to companies' ability to pursue the United Nations' SDGs. The obtained results underscore the need for managers to strategically position new technologies to foster synergies between the digital and sustainability spheres. Additionally, given the nature of the variables used in this study, the findings underscore the need for organizations to enhance their disclosure concerning AI and to leverage such tools to improve the quality and effectiveness of non-financial reporting practices. However, given the nature of AI instruments and their requirement for computational power, managers should consider the numerous organizational and operational challenges associated with AI. For instance, reliance on AI models is often tied to large energy and water consumption (Brevini, 2020; Kaplan and Haenlein, 2019; Kaplan and Haenlein, 2020), thereby underscoring the challenges inherent in the proposed relationship between organizations' AI focus and their ESG score. This manuscript underscores the social construction of technology, thus informing managers of AI instruments' logics and their connection with other elements of organizational performance.

6.3. Limitations and future research avenues

The present study carries several limitations that need to be addressed and could offer interesting future research directions. Specifically, the manuscript is limited by its geographic scope and the time period employed for data collection. All selected organizations operate within the European capital market; thus future research should explore other geographical and normative contexts to test whether different markets demonstrate different levels of interest in and sensitivity to firms' AI focus. Additionally, given the manuscript's geographical focus, it would be valuable to test the proposed hypotheses regarding organizations' AI focus and ESG scores in countries that lack a strong institutional push to promote the widespread adoption of sustainability practices among firms. A similar approach should be employed to test the obtained results in the geographical context of a developing country, where investment, digital infrastructure, and skilled labour may not be readily available. Doing so would enable scholars to better understand whether certain factors moderate or mediate the proposed relationships.

Second, the study is limited by its theoretical underpinning. Therefore, future scholars might seek to expand or shift the theoretical scope of this manuscript in an attempt to deepen our current understanding of the impact of AI on organizational performance.

Third, the study is limited by its use of LSEG Data and Analytics' ESG variable. Accordingly, researchers might test the above models by employing different measures of sustainability.

Fourth, this manuscript is subject to the methodological limitations typically associated with quantitative approaches. To further explore this line of research, scholars may consider employing qualitative methodologies, such as case studies. Additionally, future research could examine firm-level investments in digital tools like AI, as well as hiring data, to better understand organizations' transitions toward AI-driven practices and their potential impact on sustainability performance.

Finally, the constructed AI variable presents certain limitations, as it relies on data derived from organizations' narrative disclosures. While the methodological approach of building variables from corporate reporting and disclosure practices is widely accepted in the scholarly literature, future researchers may wish to consider developing a variable that allows for a more granular examination of organizations' focus on AI.

CRediT authorship contribution statement

Daniele Giordino: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Elisa Ballesio:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Data curation. **Nourah Alshaghdali:** Writing – review & editing, Supervision, Project administration. **Dhruv Galgotia:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors have no conflicting interest to declare.

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The authors have employed the use of artificial intelligence and AI-assisted tools in the writing process to improve the readability and language of this manuscript. However, it is important to note that the use of the foregoing technology has been applied with human oversight and control since AI can be incorrect, incomplete or biased. Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R543), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Appendix A

The following table details all terms employed to gather insights concerning organizations' AI focus. The following terms have been developed and detailed by Mishra et al. (2022).

Table 1
Terms Concerning Organizations' AI Focus (Obtained from Mishra et al., 2022).

Pattern recognition	Robotic process automation	Predictive analytics	Data pipelines	MathWorks
Speech recognition	Robotic	Cognitive framework	Python	Databricks
OCR	Robot	Cognitive services	TensorFlow	H20.ai
Handwriting recognition	Artificial Intelligence	Machine learning model	Google cloud platform	Datarobot
Facial recognition	Deep learning	Machine learning framework	Tensor	Amazon Machine Learning
Natural Language Processing	Neural network	Model training	Tensor Processing Unit	Google Cloud Machine Learning
Natural Language Understanding	Artificial neural network	Customer segmentation	TPU	IBM Watson
Intent classification	Convolutional neural network	Personalization engines	Cuda	Microsoft Azure Machine Learning
Slot filling	Recurrent neural network	Inferencing	Dialogflow	Azure cognitive toolkit
Entity recognition	Generative adversarial network	Embedding	Word2Vec	Apache SystemML
Semantic translation	Feedforward network	Algorithm	Doc2Vec	Apache Spark MLlib
Machine translation	Machine learning	Automation	GloVe	Apache Mahout
Chatbot	Supervised machine learning	Data science	Universal Sentence Encoding	Caffe
Autonomous agent	Unsupervised machine learning	Data acquisition	Embeddings from language models	OpenNN
Language identification	Supervised learning	Data processing	Neural network language model	Torch
Named entity extraction	Unsupervised learning	Modelling	Latent semantic analysis	Deeplearning4j
Named entity recognition	Reinforcement learning	AI Operations	Vector space models	Veles
Relationship extraction	Model registry	AIOps	Deep averaging network	Theano
Terminology extraction	Model manifestation	Machine Learning Ops	Prediction	Scikit learning
Semantic web	Model servicing	MLOps	Clustering engine	PyTorch
Computer vision	Model monitoring	Machine Learning ontologies	Topic modelling	Keras
Object recognition	Model validation	AI ethics	RapidMiner	Pandas
Intelligent word recognition	Transfer learning	Machine Learning bias	KNIME	
Intelligent image analysis	Oneshot learning	Machine bias	Alteryx	
Image processing	Pooling	Training data	SAS	

Table 2
Validation tests concerning the AI variable.

Quintile	R&D	p-value	SG&A	p-value
1	0.046		0.262	
2	0.055		0.319	
3	0.056		0.339	
4	0.069		0.372	
5	0.091		0.407	

Mean Differences (*t*-tests) Between Quintiles

Comparison	R&D t-stat	R&D p-value	SG&A t-stat	SG&A p-value
Quintile 1 vs 2	1.177	0.739	−3.975	0.000
Quintile 1 vs 3	3.466	0.790	−5.471	0.001
Quintile 1 vs 4	−2.188	0.010	−7.681	0.000
Quintile 1 vs 5	−7.899	0.000	−9.553	0.000
Quintile 2 vs 3	0.746	0.763	−1.449	0.067
Quintile 2 vs 4	−2.701	0.003	−3.870	0.000
Quintile 2 vs 5	−8.437	0.000	−6.336	0.000
Quintile 3 vs 4	−3.509	0.000	−2.302	0.003
Quintile 3 vs 5	−9.601	0.000	−4.883	0.000
Quintile 4 vs 5	−4.997	0.000	−2.116	0.009

Note: R&D and SG&A values are normalized by total assets to ensure comparability across firms. This table presents the results concerning the validation test of the AI variable. Observations gathered from 2015 to 2023 were employed.

The following table details the percentage out of the 122 terms indicated in Table 2 of AI terms used divided by year and industry.

Table 3
Percentage of AI related terms divided by year and industry.

Year	Consumer Discretionary	Communication Services	Consumer Staples	Energy	Health Care	Industrials	Information	Technology	Materials	Real Estate	Utilities
2015	8 %	9 %	10 %	8 %	18 %	12 %	10 %	27 %	15 %	12 %	10 %
2016	9 %	9 %	10 %	8 %	18 %	12 %	11 %	27 %	17 %	12 %	10 %
2017	9 %	9 %	9 %	9 %	19 %	12 %	9 %	28 %	16 %	12 %	11 %
2018	11 %	11 %	11 %	10 %	22 %	14 %	9 %	31 %	15 %	12 %	9 %
2019	10 %	10 %	11 %	10 %	21 %	16 %	11 %	30 %	18 %	14 %	10 %
2020	12 %	10 %	13 %	10 %	20 %	17 %	10 %	34 %	20 %	17 %	9 %
2021	12 %	11 %	14 %	12 %	24 %	20 %	11 %	39 %	21 %	14 %	11 %
2022	13 %	14 %	18 %	13 %	23 %	22 %	15 %	46 %	26 %	17 %	11 %
2023	12 %	14 %	17 %	13 %	24 %	24 %	18 %	48 %	25 %	20 %	13 %

Table 4
Balance test conducted on the matched sample.

	Covariance balance test of the matched sample				
	Control (Control_dummy = 0)		Treatment (Control_dummy = 1)		
Variables	Number of Obs.	Mean	Number of Obs	Mean	Mean Diff
AI	*1728*	*0.0607*	*1728*	*0.0620*	*0.0013*
Size	*1728*	*18.2129*	*1728*	*18.4403*	*0.2274*
PPE	*1728*	*0.4843*	*1728*	*0.4408*	*−0.0435*
Debt	*1728*	*0.3738*	*1728*	*0.3977*	*0.0239*
Liq	*1728*	*0.1995*	*1728*	*0.2026*	*0.0831*
RD	*1728*	*0.2887*	*1728*	*0.2978*	*0.0091*
Beta	*1728*	*0.6903*	*1728*	*0.6751*	*−0.0152*
BoardGender	*1728*	*17.0117*	*1728*	*17.2378*	*0.2261*
BoardSize	*1728*	*11.4693*	*1728*	*11.6055*	*0.1362*

This table presents the results from a balanced test conducted on the matched sample. Observations gathered from 2015 to 2023 were employed.

Data availability

Data will be made available on request.

References

Abrokwah-Larbi, K., Awuku-Larbi, Y., 2024. The impact of artificial intelligence in marketing on the performance of business organizations: evidence from SMEs in an emerging economy. Journal of Entrepreneurship in Emerging Economies 16 (4), 1090–1117. https://doi.org/10.1108/jeee-07-2022-0207.

Adobor, H., Awudu, I., Norbis, M., 2021. Integrating artificial intelligence into supply chain management: promise, challenges, and guidelines. International Journal of Logistics Systems and Management 1 (1), 1. https://doi.org/10.1504/ijlsm.2021.10038691.

Agrawal, A., Gans, J., Goldfarb, A., 2019a. The Economics of Artificial Intelligence: An Agenda. University of Chicago Press.

Agrawal, Ajay, Gans, J.S., Goldfarb, A., 2019b. Artificial intelligence: the ambiguous labor market impact of automating prediction. The Journal of Economic Perspectives: A Journal of the American Economic Association 33 (2), 31–50. https://doi.org/10.1257/jep.33.2.31.

Ahdadou, M., Aajly, A., Tahrouch, M., 2024. Enhancing corporate governance through AI: a systematic literature review. Tech. Anal. Strat. Manag. 1–14. https://doi.org/10.1080/09537325.2024.2326120.

Ahmad, T., Zhang, D., Huang, C., Zhang, H., Dai, N., Song, Y., Chen, H., 2021. Artificial intelligence in sustainable energy industry: status quo, challenges and opportunities. J. Clean. Prod. 289, 125834. https://doi.org/10.1016/j.jclepro.2021.125834.

Akter, S., Babu, M.M., Hani, U., Sultana, S., Bandara, R., Grant, D., 2024. Unleashing the power of artificial intelligence for climate action in industrial markets. Ind. Mark. Manag. 117, 92–113. https://doi.org/10.1016/j.indmarman.2023.12.011.

Ameye, N., Bughin, J., van Zeebroeck, N., 2023. How uncertainty shapes herding in the corporate use of artificial intelligence technology. Technovation 127, 102846. https://doi.org/10.1016/j.technovation.2023.102846.

Appio, F.P., Cacciatore, E., Cesaroni, F., Crupi, A., Marozzo, V., 2024. Open innovation at the digital frontier: unraveling the paradoxes and roadmaps for SMEs' successful digital transformation. Eur. J. Innov. Manag. 27 (9), 223–247. https://doi.org/10.1108/ejim-04-2023-0343.

Azadi, M., Yousefi, S., Farzipoor Saen, R., Shabanpour, H., Jabeen, F., 2023. Forecasting sustainability of healthcare supply chains using deep learning and network data envelopment analysis. J. Bus. Res. 154, 113357. https://doi.org/10.1016/j.jbusres.2022.113357.

Bekaert, G., Rothenberg, R., Noguer, M., 2023. Sustainable investment-exploring the linkage between alpha, ESG, and SDGs. Sustain. Dev. 5, 3831–3842. https://doi.org/10.1002/sd.2628.

Berger, A.N., Kick, T., Schaeck, K., 2014. Executive board composition and bank risk taking. Finance 28, 48–65. https://doi.org/10.1016/j.jcorpfin.2013.11.006.

Bibri, S.E., Krogstie, J., Kaboli, A., Alahi, A., 2024. Smarter eco-cities and their leading-edge artificial intelligence of things solutions for environmental sustainability: a comprehensive systematic review. Environmental Science and Ecotechnology 19, 100330. https://doi.org/10.1016/j.ese.2023.100330.

Bock, D.E., Wolter, J.S., Ferrell, O.C., 2020. Artificial intelligence: disrupting what we know about services. J. Serv. Mark. 34 (3), 317–334. https://doi.org/10.1108/jsm-01-2019-0047.

Brevini, B., 2020. Black boxes, not green: mythologizing artificial intelligence and omitting the environment. Big Data Soc. 7 (2), 205395172093514. https://doi.org/10.1177/2053951720935141.

Buehlmaier, M.M.M., Whited, T.M., 2018. Are financial constraints priced? Evidence from textual analysis. Rev. Financ. Stud. 31 (7), 2693–2728. https://doi.org/10.1093/rfs/hhy007.

Cao, G., Duan, Y., Edwards, J.S., Dwivedi, Y.K., 2021. Understanding managers' attitudes and behavioral intentions towards using artificial intelligence for organizational decision-making. Technovation 106 (102312), 102312. https://doi.org/10.1016/j.technovation.2021.102312.

Chhillar, D., Aguilera, R.V., 2022. An eye for artificial intelligence: insights into the governance of artificial intelligence and vision for future research. Bus. Soc. 61 (5), 1197–1241. https://doi.org/10.1177/00076503221080959.

Chiarello, F., Giordano, V., Spada, I., Barandoni, S., Fantoni, G., 2024. Future applications of generative large language models: a data-driven case study on ChatGPT. Technovation 133 (103002), 103002. https://doi.org/10.1016/j.technovation.2024.103002.

Choi, J., Wang, H., 2009. Stakeholder relations and the persistence of corporate financial performance. Strat. Manage. J. 30 (8), 895–907. https://doi.org/10.1002/smj.759.

Corazza, M., Garcia, R., Khan, F.S., Torre, L., 2024. Artificial intelligence and beyond for Finance, 15. World scientific.

Courtland, R., 2018. Bias detectives: the researchers striving to make algorithms fair. Nature 558 (7710), 357–360. https://doi.org/10.1038/d41586-018-05469-3.

Damoah, I.S., Ayakwah, A., Tingbani, I., 2021. Artificial intelligence (AI)-enhanced medical drones in the healthcare supply chain (HSC) for sustainability development: a case study. J. Clean. Prod. 328, 129598. https://doi.org/10.1016/j.jclepro.2021.129598.

Davenport, T.H., Ronanki, R., 2018. Artificial intelligence for the real world. Harv. Bus. Rev. 96 (1), 108–116.

Delgado-Ceballos, J., Ortiz-De-Mandojana, N., Antolín-López, R., Montiel, I., 2023. Connecting the sustainable development goals to firm-level sustainability and ESG factors: the need for double materiality. BRQ Bus. Res. Q. 26 (1), 2–10. https://doi.org/10.1177/23409444221140919.

Dobrick, J., Klein, C., Zwergel, B., 2023. Size bias in refinitiv ESG data. Financ. Res. Lett. 104014, 104014. https://doi.org/10.1016/j.frl.2023.104014.

Du, S., Xie, C., 2021. Paradoxes of artificial intelligence in consumer markets: ethical challenges and opportunities. J. Bus. Res. 129, 961–974. https://doi.org/10.1016/j.jbusres.2020.08.024.

Dwivedi, Y.K., Sharma, A., Rana, N.P., Giannakis, M., Goel, P., Dutot, V., 2023. Evolution of artificial intelligence research in technological forecasting and social change: research topics, trends, and future directions. Technol. Forecast. Soc. Chang. 192, 122579. https://doi.org/10.1016/j.techfore.2023.122579.

Emma, G.-M., Jennifer, M.-F., 2021. Is SDG reporting substantial or symbolic? An examination of controversial and environmentally sensitive industries. J. Clean. Prod. 298 (126781), 126781. https://doi.org/10.1016/j.jclepro.2021.126781.

Enholm, I.M., Papagiannidis, E., Mikalef, P., Krogstie, J., 2022. Artificial intelligence and business value: a literature review. Information Systems Frontiers: A Journal of Research and Innovation 24 (5), 1709–1734. https://doi.org/10.1007/s10796-021-10186-w.

Erhart, S., 2022. Take it with a pinch of salt-ESG rating of stocks and stock indices. Int. Rev. Financ. Anal. 83 (102308). https://doi.org/10.1016/j.irfa.2022.102308.

Ertemel, A.V., Karadayı, T., Makaritou, P., 2021. Investigating the socio-economic consequences of artificial intelligence: *a qualitative research*. Journal of International Trade, Logistics and Law 7 (1), 75–89.

Ertz, M., Sun, S., Boily, E., Kubiat, P., Quenum, G.G.Y., 2022. How transitioning to industry 4.0 promotes circular product lifetimes. Ind. Mark. Manag. 101, 125–140. https://doi.org/10.1016/j.indmarman.2021.11.014.

Far, S.B., Rad, A.I., Asaar, M.R., 2023. Blockchain and its derived technologies shape the future generation of digital businesses: a focus on decentralized finance and the Metaverse. Data Science and Management 6 (3), 183–197. https://doi.org/10.1016/j.dsm.2023.06.002.

Feuerriegel, S., Hartmann, J., Janiesch, C., Zschech, P., 2023. Generative AI. SSRN Electron. J. 66 (1), 111–126. https://doi.org/10.2139/ssrn.4443189.

Frasca, M., La Torre, D., Pravettoni, G., Cutica, I., 2024. Explainable and interpretable artificial intelligence in medicine: a systematic bibliometric review. Discov. Artif. Intell. 4 (1), 15. https://doi.org/10.1007/s44163-024-00114-7.

Ghosh, V., Chotia, V., Alghafes, R., Basahel, S., Nazrul, A., 2025. Bridging AI minds and corporate sustainability-exploring the role of business model innovation and organizational sensemaking. Technol. Forecast. Soc. Chang. 218, 124229. https://doi.org/10.1016/j.techfore.2025.124229.

Giarmoleo, F.V., Ferrero, I., Rocchi, M., Pellegrini, M.M., 2024. What ethics can say on artificial intelligence: insights from a systematic literature review. Bus. Soc. Rev. 129 (2), 258–292. https://doi.org/10.1111/basr.12336.

Gill, S.S., Xu, M., Ottaviani, C., Patros, P., Bahsoon, R., Shaghaghi, A., Golec, M., Stankovski, V., Wu, H., Abraham, A., Singh, M., Mehta, H., Ghosh, S.K., Baker, T., Parlikad, A.K., Lutfiyya, H., Kanhere, S.S., Sakellariou, R., Dustdar, S., Uhlig, S., 2022. AI for next generation computing: emerging trends and future directions. Internet of Things 19, 100514. https://doi.org/10.1016/j.iot.2022.100514.

Giordano, Vito, Fantoni, G., 2025. Decomposing maintenance actions into sub-tasks using natural language processing: a case study in an Italian automotive company. Comput. Ind. 164 (104186), 104186. https://doi.org/10.1016/j.compind.2024.104186.

Giordano, Vito, Spada, I., Chiarello, F., Fantoni, G., 2024. The impact of ChatGPT on human skills: a quantitative study on twitter data. Technol. Forecast. Soc. Chang. 203, 123389. https://doi.org/10.1016/j.techfore.2024.123389.

Giordino, D., Ballesio, E., Galgotia, A., Broccardo, L., 2025a. Does institutional ownership structure reduce greenhouse gas emissions? An in-depth study of corporations social responsibility of European-listed firms. Corp. Soc. Responsib. Environ. Manag. https://doi.org/10.1002/csr.70219 ahead-of-print.

Giordino, D., Jabeen, F., Nirino, N., Bresciani, S., 2024. Institutional investors ownership concentration and its effect on disclosure and transparency of United Nations sustainable development goals. Technol. Forecast. Soc. Chang. 200, 123132. https://doi.org/10.1016/j.techfore.2023.123132.

Giordino, D., Sestino, A., Zàmek, D., Yahiaoui, D., 2025b. The impact of institutional ownership structure on corporate ESG performance: empirical evidence for the European capital market. Rev. Manag. Sci. https://doi.org/10.1007/s11846-025-00879-w ahead-of-print.

Gupta, R., Nair, K., Mishra, M., Ibrahim, B., Bhardwaj, S., 2024. Adoption and impacts of generative artificial intelligence: theoretical underpinnings and research agenda. International Journal of Information Management Data Insights 4 (1). https://doi.org/10.1016/j.jjimei.2024.100232.

Gupta, Shaphali, Leszkiewicz, A., Kumar, V., Bijmolt, T., Potapov, D., 2020. Digital analytics: modeling for insights and new methods. J. Interact. Mark. 51 (1), 26–43. https://doi.org/10.1016/j.intmar.2020.04.003.

Gupta, Shivam, Langhans, S.D., Domisch, S., Fuso-Nerini, F., Felländer, A., Battaglini, M., Tegmark, M., Vinuesa, R., 2021. Assessing whether artificial intelligence is an enabler or an inhibitor of sustainability at indicator level. Transp. Eng. Aust. 4, 100064. https://doi.org/10.1016/j.treng.2021.100064.

Gupta, Shivam, Modgil, S., Choi, T.-M., Kumar, A., Antony, J., 2023. Influences of artificial intelligence and blockchain technology on financial resilience of supply chains. Int. J. Prod. Econ. 261, 108868. https://doi.org/10.1016/j.ijpe.2023.108868.

Haefner, N., Parida, V., Gassmann, O., Wincent, J., 2023. Implementing and scaling artificial intelligence: a review, framework, and research agenda. Technol. Forecast. Soc. Chang. 197, 122878. https://doi.org/10.1016/j.techfore.2023.122878.

Hair, J.F., 2011. Multivariate data analysis: An overview. In: International Encyclopedia of Statistical Science. Springer Berlin Heidelberg, pp. 904–907. https://doi.org/10.1007/978-3-642-04898-2_395.

Hang, H., Chen, Z., 2022. How to realize the full potentials of artificial intelligence (AI) in digital economy? A literature review. Journal of Digital Economy 1 (3), 180–191. https://doi.org/10.1016/j.jdec.2022.11.003.

Heilinger, J.-C., Kempt, H., Nagel, S., 2024. Beware of sustainable AI! Uses and abuses of a worthy goal. AI and Ethics 4 (2), 201–212. https://doi.org/10.1007/s43681-023-00259-8.

Holmström, J., Carroll, N., 2024. How organizations can innovate with generative AI. Bus. Horiz. 68 (5), 559–573. https://doi.org/10.1016/j.bushor.2024.02.010.

Huang, M.-H., Rust, R.T., 2021. A strategic framework for artificial intelligence in marketing. J. Acad. Mark. Sci. 49 (1), 30–50. https://doi.org/10.1007/s11747-020-00749-9.

Huang, X., Yang, F., Zheng, J., Feng, C., Zhang, L., 2023. Personalized human resource management via HR analytics and artificial intelligence: theory and implications. Asia Pac. Manag. Rev. 28 (4), 598–610. https://doi.org/10.1016/j.apmrv.2023.04.004.

Hudson, K., Morgan, R.E., 2024. Industry exposure to artificial intelligence, board network heterogeneity, and firm idiosyncratic risk. J. Manag. Stud. https://doi.org/10.1111/joms.13127.

Jammeli, H., Ksantini, R., Ben Abdelaziz, F., Masri, H., 2023. Sequential artificial intelligence models to forecast urban solid waste in the city of Sousse, Tunisia. IEEE Trans. Eng. Manag. 70 (5), 1912–1922. https://doi.org/10.1109/tem.2021.3081609.

John-Mathews, J.-M., 2022. Some critical and ethical perspectives on the empirical turn of AI interpretability. Technol. Forecast. Soc. Chang. 174 (121209), 121209. https://doi.org/10.1016/j.techfore.2021.121209.

de Jong, M.D.T., van der Meer, M., 2017. How does it fit? Exploring the congruence between organizations and their corporate social responsibility (CSR) activities. J. Bus. Ethics 143 (1), 71–83. https://doi.org/10.1007/s10551-015-2782-2.

Kaplan, A., Haenlein, M., 2019. Siri, Siri, in my hand: who's the fairest in the land? On the interpretations, illustrations, and impli- cations of artificial intelligence. Bus. Horiz. 62 (1), 15–25. https://doi.org/10.1016/j.bushor.2018.08.004.

Kaplan, Andreas, Haenlein, M., 2020. Rulers of the world, unite! The challenges and opportunities of artificial intelligence. Bus. Horiz. 63 (1), 37–50. https://doi.org/10.1016/j.bushor.2019.09.003.

Kar, A.K., Choudhary, S.K., Singh, V.K., 2022. How can artificial intelligence impact sustainability: a systematic literature review. J. Clean. Prod. 376 (134120), 134120. https://doi.org/10.1016/j.jclepro.2022.134120.

Khakurel, J., Penzenstadler, B., Porras, J., Knutas, A., Zhang, W., 2018. The rise of artificial intelligence under the lens of sustainability. Technologies 6 (4), 100. https://doi.org/10.3390/technologies6040100.

Kogut, B., Zander, U., 1992. Knowledge of the firm, combinative capabilities, and the replication of technology. Organ. Sci. 3 (3), 383–397. https://doi.org/10.1287/orsc.3.3.383.

Kulkov, I., Kulkova, J., Rohrbeck, R., Menvielle, L., Kaartemo, V., Makkonen, H., 2024. Artificial intelligence-driven sustainable development: examining organizational, technical, and processing approaches to achieving global goals. Sustain. Dev. 32 (3), 2253–2267. https://doi.org/10.1002/sd.2773.

Kumar, A., Bhattacharyya, S.S., Krishnamoorthy, B., 2023. Automation-augmentation paradox in organizational artificial intelligence technology deployment capabilities; an empirical investigation for achieving simultaneous economic and social benefits. J. Enterp. Inf. Manag. 36 (6), 1556–1582. https://doi.org/10.1108/jeim-09-2022-0307.

La Torre, D., Appio, F., Lazzeri, F., Schiavone, F., Masri, H., 2023. Impact of artificial intelligence in business and society: opportunities and challenges. Impact of Artificial Intelligence in Business and Society 83–167.

Lanzolla, G., Lorenz, A., Miron-Spektor, E., Schilling, M., Solinas, G., Tucci, C.L., 2020. Digital transformation: what is new if anything? Emerging patterns and management research. Academy of Management Discoveries 6, 341–350. https://doi.org/10.5465/amd.2020.0144.

Ledro, C., Nosella, A., Dalla Pozza, I., 2023. Integration of AI in CRM: challenges and guidelines. J. Open Innov.: Technol. Mark. Complex. 9 (4), 100151. https://doi.org/10.1016/j.joitmc.2023.100151.

Leone, D., Schiavone, F., Appio, F.P., Chiao, B., 2021. How does artificial intelligence enable and enhance value co-creation in industrial markets? An exploratory case study in the healthcare ecosystem. J. Bus. Res. 129, 849–859. https://doi.org/10.1016/j.jbusres.2020.11.008.

Li, J., Wu, T., Hu, B., Pan, D., Zhou, Y., 2025. Artificial intelligence and corporate ESG performance. Int. Rev. Financ. Anal. 102 (104036), 104036. https://doi.org/10.1016/j.irfa.2025.104036.

Li, Y., Gong, M., Zhang, X., Koh, L., 2018. The impact of environmental, social, and governance disclosure on firm value. Br. Account. Rev. 50 (1), 60–75. https://doi.org/10.1016/j.bar.2017.09.007.

Lins, K.V., Servaes, H., Tamayo, A., 2019. Social capital, trust, and corporate performance: how CSR helped companies during the financial crisis (and why it can keep helping them). J. Appl. Corp. Financ. 31 (2), 59–71. https://doi.org/10.1111/jacf.12347.

London Stock Exchange Group (2024) ESG scores. https://www.lseg.com/en/data-analytics/sustainable-finance/esg-scores. Accessed 8 January 2024.

Madanaguli, A., Sjödin, D., Parida, V., Mikalef, P., 2024. Artificial intelligence capabilities for circular business models: research synthesis and future agenda. Technol. Forecast. Soc. Chang. 200 (123189), 123189. https://doi.org/10.1016/j.techfore.2023.123189.

Makarius, E.E., Mukherjee, D., Fox, J.D., Fox, A.K., 2020. Rising with the machines: a sociotechnical framework for bringing artificial intelligence into the organization. J. Bus. Res. 120, 262–273. https://doi.org/10.1016/j.jbusres.2020.07.045.
Mariani, M.M., Machado, I., Magrelli, V., Dwivedi, Y.K., 2023. Artificial intelligence in innovation research: a systematic review, conceptual framework, and future research directions. Technovation 122 (102623), 102623. https://doi.org/10.1016/j.technovation.2022.102623.
Marwala, T., Hurwitz, E., 2017. Artificial Intelligence and Economic Theory: Skynet in the Market, 1st ed. Springer International Publishing.
Miozza, M., Brunetta, F., Appio, F.P., 2024. Digital transformation of the pharmaceutical industry: a future research agenda for management studies. Technol. Forecast. Soc. Chang. 207 (123580), 123580. https://doi.org/10.1016/j.techfore.2024.123580.
Mishra, S., Ewing, M.T., Cooper, H.B., 2022. Artificial intelligence focus and firm performance. J. Acad. Mark. Sci. 50 (6), 1176–1197. https://doi.org/10.1007/s11747-022-00876-5.
Modgil, S., Singh, R.K., Hannibal, C., 2022. Artificial intelligence for supply chain resilience: learning from Covid-19. Int. J. Logist. Manag. 33 (4), 1246–1268. https://doi.org/10.1108/ijlm-02-2021-0094.
Monteiro, A.P., Vale, J., Leite, E., Lis, M., Kurowska-Pysz, J., 2022. The impact of information systems and non-financial information on company success. Int. J. Account. Inf. Syst. 45, 100557. https://doi.org/10.1016/j.accinf.2022.100557.
Moradi, M., Dass, M., 2022. Applications of artificial intelligence in B2B marketing: challenges and future directions. Ind. Mark. Manag. 107, 300–314. https://doi.org/10.1016/j.indmarman.2022.10.016.
Neiroukh, S., Emeagwali, O.L., Aljuhmani, H.Y., 2024. Artificial intelligence capability and organizational performance: unraveling the mediating mechanisms of decision-making processes. Manag. Decis. https://doi.org/10.1108/md-10-2023-1946 ahead-of-print.
Nishant, R., Kennedy, M., Corbett, J., 2020. Artificial intelligence for sustainability: challenges, opportunities, and a research agenda. Int. J. Inf. Manag. 53 (102104), 102104. https://doi.org/10.1016/j.ijinfomgt.2020.102104.
Olan, F., Ogiemwonyi Arakpogun, E., Suklan, J., Nakpodia, F., Damij, N., Jayawickrama, U., 2022. Artificial intelligence and knowledge sharing: contributing factors to organizational performance. J. Bus. Res. 145, 605–615. https://doi.org/10.1016/j.jbusres.2022.03.008.
Omrani, N., Rivieccio, G., Fiore, U., Schiavone, F., Agreda, S.G., 2022. To trust or not to trust? An assessment of trust in AI-based systems: concerns, ethics and contexts. Technol. Forecast. Soc. Change 181 (121763), 121763. https://doi.org/10.1016/j.techfore.2022.121763.
Pan, S.L., Nishant, R., 2023. Artificial intelligence for digital sustainability: an insight into domain-specific research and future directions. Int. J. Inf. Manag. 72 (102668), 102668. https://doi.org/10.1016/j.ijinfomgt.2023.102668.
Parteka, A., Kordalska, A., 2023. Artificial intelligence and productivity: global evidence from AI patent and bibliometric data. Technovation 125 (102764), 102764. https://doi.org/10.1016/j.technovation.2023.102764.
Peltier, J.W., Dahl, A.J., Schibrowsky, J.A., 2024. Artificial intelligence in interactive marketing: a conceptual framework and research agenda. J. Res. Interact. Mark. 18 (1), 54–90. https://doi.org/10.1108/JRIM-01-2023-0030.
Peters, D., Vold, K., Robinson, D., Calvo, R.A., 2020. Responsible AI—two frameworks for ethical design practice. IEEE Transactions on Technology and Society 1 (1), 34–47. https://doi.org/10.1109/tts.2020.2974991.
Pham, P., Zhang, H., Gao, W., Zhu, X., 2024. Determinants and performance outcomes of artificial intelligence adoption: Evidence from US hospitals. J. Bus. Res. 172. https://doi.org/10.1016/j.jbusres.2023.114402.
Rahman, M.S., Bag, S., Gupta, S., Sivarajah, U., 2023. Technology readiness of B2B firms and AI-based customer relationship management capability for enhancing social sustainability performance. J. Bus. Res. 156 (113525), 113525. https://doi.org/10.1016/j.jbusres.2022.113525.
Rammer, C., Fernández, G.P., Czarnitzki, D., 2022. Artificial intelligence and industrial innovation: evidence from German firm-level data. Res. Policy 51 (7), 104555. https://doi.org/10.1016/j.respol.2022.104555.
Rana, N.P., Chatterjee, S., Dwivedi, Y.K., Akter, S., 2022. Understanding dark side of artificial intelligence (AI) integrated business analytics: assessing firm's operational inefficiency and competitiveness. European Journal of Information Systems: An Official Journal of the Operational Research Society 31 (3), 364–387. https://doi.org/10.1080/0960085x.2021.1955628.
Roberts, D.L., Candi, M., 2024. Artificial intelligence and innovation management: charting the evolving landscape. Technovation 136 (103081), 103081. https://doi.org/10.1016/j.technovation.2024.103081.
Ronaghi, M.H., 2023. The influence of artificial intelligence adoption on circular economy practices in manufacturing industries. Environ. Dev. Sustain. 25 (12), 14355–14380. https://doi.org/10.1007/s10668-022-02670-3.
Rosenbaum, P.R., Rubin, D.B., 1983. The central role of the propensity score in observational studies for causal effects. Biometrika 70 (1), 41–55. https://doi.org/10.1093/biomet/70.1.41.
Schneider, J., Abraham, R., Meske, C., Vom Brocke, J., 2023. Artificial intelligence governance for businesses. Inf. Syst. Manag. 40 (3), 229–249. https://doi.org/10.1080/10580530.2022.2085825.
Schwaeke, J., Gerlich, C., Nguyen, H.L., Kanbach, D.K., Gast, J., 2025. Artificial intelligence (AI) for good? Enabling organizational change towards sustainability. Rev. Manag. Sci. 1–26. https://doi.org/10.1007/s11846-025-00840-x.
Shiyyab, F.S., Alzoubi, A.B., Obidat, Q.M., Alshurafat, H., 2023. The impact of artificial intelligence disclosure on financial performance. International Journal of Financial Studies 11 (3). https://doi.org/10.3390/ijfs11030115.
Singh, N., Jain, M., Kamal, M.M., Bodhi, R., Gupta, B., 2024. Technological paradoxes and artificial intelligence implementation in healthcare. An application of paradox theory. Technol. Forecast. Soc. Chang. 198 (122967), 122967. https://doi.org/10.1016/j.techfore.2023.122967.
Sipola, J., Saunila, M., Ukko, J., 2023. Adopting artificial intelligence in sustainable business. J. Clean. Prod. 426 (139197), 139197. https://doi.org/10.1016/j.jclepro.2023.139197.
Sjödin, D., Parida, V., Kohtamäki, M., 2023. Artificial intelligence enabling circular business model innovation in digital servitization: conceptualizing dynamic capabilities, AI capacities, business models and effects. Technol. Forecast. Soc. Chang. 197 (122903), 122903. https://doi.org/10.1016/j.techfore.2023.122903.
Spring, M., Faulconbridge, J., Sarwar, A., 2022. How information technology automates and augments processes: insights from artificial-intelligence-based systems in professional service operations. J. Oper. Manag. 68 (6–7), 592–618. https://doi.org/10.1002/joom.1215.
Stahl, B.C., Andreou, A., Brey, P., Hatzakis, T., Kirichenko, A., Macnish, K., Wright., 2021. Artificial intelligence for human flourishing-beyond principles for machine learning. J. Bus. Res. 124, 374–388. https://doi.org/10.1016/j.jbusres.2020.11.030.
Starks, L.T., 2023. Presidential address: sustainable finance and ESG issues—valueversusvalues. J. Financ. 78 (4), 1837–1872. https://doi.org/10.1111/jofi.13255.
Tian, H., Zhao, L., Yunfang, L., Wang, W., 2023. Can enterprise green technology innovation performance achieve "corner overtaking" by using artificial intelligence?-evidence from Chinese manufacturing enterprises. Technol. Forecast. Soc. Chang. 194. https://doi.org/10.1016/j.techfore.2023.122732.
Tingbani, I., Salia, S., Hartwell, C.A., Yahaya, A., 2024. Looking in the rear-view mirror: evidence from artificial intelligence investment, labour market conditions and firm growth. Int. J. Financ. Econ. 30 (1), 961–982. https://doi.org/10.1002/ijfe.2945.
Toorajipour, R., Sohrabpour, V., Nazarpour, A., Oghazi, P., Fischl, M., 2021. Artificial intelligence in supply chain management: a systematic literature review. J. Bus. Res. 122, 502–517. https://doi.org/10.1016/j.jbusres.2020.09.009.
Triberti, S., Durosini, I., La Torre, D., Sebri, V., Savioni, L., Pravettoni, G., 2022. Artificial intelligence in healthcare practice: How to tackle the "human" challenge. In: Handbook of Artificial Intelligence in Healthcare. Springer International Publishing, pp. 43–60.
Truant, E., Giordino, D., Borlatto, E., Bhatia, M., 2024. Drivers and barriers of smart technologies for circular economy: leveraging smart circular economy implementation to nurture companies' performance. Technol. Forecast. Soc. Chang. 198 (122954), 122954. https://doi.org/10.1016/j.techfore.2023.122954.
Vaio, D., Palladino, A., Hassan, R., Escobar, R., 2020. Artificial intelligence and business models in the sustainable development goals perspective: a systematic literature review. J. Bus. Res. 121, 283–314. https://doi.org/10.1016/j.jbusres.2020.08.019.
Velte, P., 2017. Does ESG performance have an impact on financial performance? Evidence from Germany. Journal of Global Responsibility 8 (2), 169–178. https://doi.org/10.1108/jgr-11-2016-0029.
Velte, P., 2023. Which institutional investors drive corporate sustainability? A systematic literature review. Bus. Strateg. Environ. 32 (1), 42–71. https://doi.org/10.1002/bse.3117.
Vinuesa, R., Azizpour, H., Leite, I., Balaam, M., Dignum, V., Domisch, S., Felländer, A., Langhans, S.D., Tegmark, M., Fuso Nerini, F., 2020. The role of artificial intelligence in achieving the sustainable development goals. Nat. Commun. 11 (1), 233. https://doi.org/10.1038/s41467-019-14108-y.
Vlačić, B., Corbo, L., Costa e Silva, S., Dabić, M., 2021. The evolving role of artificial intelligence in marketing: a review and research agenda. J. Bus. Res. 128, 187–203. https://doi.org/10.1016/j.jbusres.2021.01.055.
Wamba-Taguimdje, S.-L., Fosso Wamba, S., Kala Kamdjoug, J.R., Tchatchouang Wanko, C.E., 2020. Influence of artificial intelligence (AI) on firm performance: the business value of AI-based transformation projects. Bus. Process. Manag. J. 26 (7), 1893–1924. https://doi.org/10.1108/bpmj-10-2019-0411.
Wang, S., Zhang, Y., Xiao, Y., Liang, Z., 2025. Artificial intelligence policy frameworks in China, the European Union and the United States: an analysis based on structure topic model. Technol. Forecast. Soc. Chang. 212 (123971), 123971. https://doi.org/10.1016/j.techfore.2025.123971.
Wang, Z., Zhang, T., Ren, X., Shi, Y., 2024. AI adoption rate and corporate green innovation efficiency: evidence from Chinese energy companies. Energy Econ. 132 (107499), 107499. https://doi.org/10.1016/j.eneco.2024.107499.
Wirtz, B.W., Weyerer, J.C., Sturm, B.J., 2020. The dark sides of artificial intelligence: an integrated AI governance framework for public administration. Int. J. Public Adm. 43 (9), 818–829. https://doi.org/10.1080/01900692.2020.1749851.
Wooldridge, J.M., 2009. Chapter 6-Multiple Regression Analysis: Further Issues. In: Introductory Econometrics: A Modern Approach, fifth 881. Thomson-South Western, US.
Xiao, Y., Xiang, R., Sun, Y.-L., Chen, J., Hao, Y.-H., 2025. Digital disruption, knowledge and collaborative networks and green innovation in China manufacturing transformation. Technol. Forecast. Soc. Chang. 216 (124120), 124120. https://doi.org/10.1016/j.techfore.2025.124120.
Yu, X., Xu, S., Ashton, M., 2023. Antecedents and outcomes of artificial intelligence adoption and application in the workplace: the socio-technical system theory perspective. Inf. Technol. People 36 (1), 454–474. https://doi.org/10.1108/itp-04-2021-0254.
Zeira, J., 1998. Workers, machines, and economic growth. The quarterly. J. Econ. 113 (4), 1091–1117. https://doi.org/10.1162/003355398555847.

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